STRATA: One Geometry-Free Persistence-and-Periodicity Engine for Selectable 2D/3D Lifelong LiDAR Mapping

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A robot that maps one building for weeks must keep walls (durability), forget movers (plasticity) and treat a scheduled door as signal, not noise (periodicity). Per-cell persistence and periodicity models already exist; the closest precedent is Frequency Map Enhancement (FreMEn), in the 2D grid of Krajník et al. (2016) and in FROctomap. We present *STRATA*, a SLAM-free engine that combines known ingredients in one per-cell state machine—windowed log-odds with survival decay, a Schmitt-trigger graduation band, pruning, and a FreMEn-style Fourier test at one configured base period and its harmonics, without period selection—shared unchanged by a fixed-array 2D grid and a sparse 3D voxel hash that differ only in the point-to-key map and the ray walk. Standard inequalities give the periodic label a false-alarm bound for iid clutter, with the level spent over the touch count; it holds per cell history, so only for unpruned cells, and pruned clutter is measured. All experiments use hand-authored synthetic hit and miss patterns; no real or simulated LiDAR scan is processed, and no external baseline, FreMEn included, is run. Both backends graduate a static wall by the third window and keep recall 1.0; removing hysteresis raises flicker from 54 to 574 toggles. With the level fixed on calibration seeds, at most 2 of 2000 Bernoulli clutter cells are ever labelled Periodic within 1024 windows in every held-out configuration (Wilson 95% upper limit 0.0036; the proved bound is 0.2 per unpruned history), and 50%-duty doors are detected from 25 windows (8 without the test). The dominant error is a false Static label: with the shipped defaults 1960 of 2000 Bernoulli(0.5) clutter cells end Static, because the classifier cannot tell random clutter from a noisy wall of similar occupancy. At the shipped period, temporally correlated clutter is labelled Periodic. Code: `https://github.com/kjungmo/strata`.

Keywords: lifelong mapping, occupancy grids, voxel hashing, persistence filtering, FreMEn periodicity, ROS 2

1. Introduction

An occupancy grid built the textbook way accumulates evidence and never lets go of it (Thrun et al., 2005). For a robot that drives a route once this is harmless; for one that maps the same warehouse or hospital corridor for weeks it is a slow failure. A cart parked for an afternoon, a person walking past and a load-bearing wall all leave the same kind of mark, and after enough passes the map describes the union of everything the sensor ever touched rather than the building. Lifelong mapping has to hold three requirements at once: the map must be *durable* enough to keep structure that is genuinely permanent, *plastic* enough to forget structure that has moved on, and it must do both without one global forgetting rate that is simultaneously too slow to erase movers and too fast to trust a wall—the stability–plasticity dilemma that Biber and Duckett (2005) framed for dynamic maps. A third requirement complicates the split: some structure is *cyclic*, such as a door open every morning

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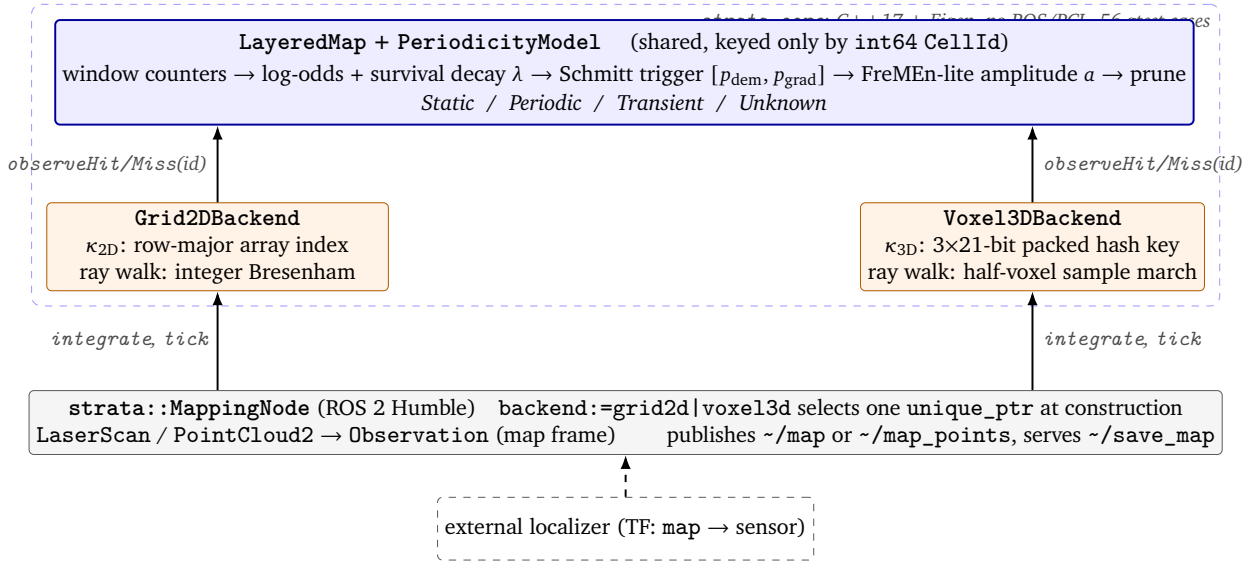


Figure 1: Overview of STRATA. One geometry-free engine (top) owns every persistence and periodicity decision and sees only integer cell keys. The two backends (middle) contribute exactly two things: the point-to-key map κ_b and the free-space ray walk. A thin ROS 2 node (bottom) selects one backend from a single string parameter and consumes, but never produces, the sensor pose.

and shut every night. Its occupancy is a periodic temporal signal, and a flat decay law erases exactly the regularity that makes it predictable (Krajník et al., 2014a, 2017).

Per-cell models of these requirements are not new. Temporal occupancy grids (Arbuckle et al., 2002) classify cells by their occupancy over several timescales and use the result to locate doors, and Mitsou and Tzafestas (2007) keep each cell's occupancy history to separate static, low-dynamic and high-dynamic cells. The closest prior work is the FreMEn line. Krajník et al. (2016) maintain a 2D spatio-temporal occupancy grid that represents the persistence and periodicity of individual cells; it integrates maps that gmapping builds with AMCL pose estimates as odometry, replaces the map server of the ROS navigation stack, and was evaluated on data from several days of routine autonomous patrols of an open-plan office. FROctomap (Krajník et al., 2014b) attaches the same spectral model to the voxels of a 3D octree. Several recent LiDAR lifelong-mapping systems instead place persistence logic inside larger pipelines. ELite (Gil et al., 2025) and LT-mapper (Kim and Kim, 2022) resolve durability against plasticity as stages of full lifelong-SLAM pipelines with multi-session registration, place recognition and alignment. RTAB-Map (Labbé and Michaud, 2019) offers selectable 2D and 3D geometry under one framework, as a complete SLAM system with loop closure and memory tiering. Khronos (Schmid et al., 2024) generalises the two-timescale idea to a dense spatio-temporal metric-semantic map, and MTD-Map (Kim et al., 2026), concurrent with this work, maintains static, dynamic and transition maps from per-voxel statistics of occupancy transitions. Every ingredient STRATA uses therefore has precedent. What we did not find in this literature is their particular combination: one small, SLAM-free implementation that assigns each cell a discrete label through hysteresis graduation and pruning, is shared unchanged by a 2D grid and a 3D voxel hash, and bounds the false-alarm probability of its periodic label in finite samples under an iid null.

We present *STRATA*, a persistence-and-periodicity engine whose cells carry temporal strata—Static, Periodic and Transient classes over an Unknown baseline—inside one per-cell state machine (Figure 1). The state machine never sees coordinates. Both shipped backends translate world points into `int64` cell keys and walk a free-space ray to clear it; everything downstream of the key happens once, in the shared engine. Choosing 2D or 3D is one runtime string parameter resolved at construction, and

the engine (`strata_core`) links no ROS, DDS, TF or PCL, so its scientific logic builds and unit-tests without a ROS installation. We state the scope up front: STRATA is not SLAM (it consumes an external `map`→sensor transform), it models single-session persistence rather than multi-session remapping, its periodicity test checks one configured base period and its harmonics rather than selecting periods from a spectrum as FreMEn does, and its evaluation is synthetic, with no external baseline. Our contributions are:

- *One Geometry-Free Engine, Two Geometries*: a single `int64`-keyed engine (`LayeredMap` with `PeriodicityModel`) drives a fixed-array 2D grid and a sparse 3D voxel hash through one `MapBackend` interface; the only per-backend code is the point-to-key map and the free-space ray walk (§3), and the core builds with C++17 and Eigen alone under 56 behaviour-level tests. Per-cell periodic grids already exist in 2D and in 3D (Krajník et al., 2014b, 2016), as separate implementations (STRANDS Project, 2017); what is added here is one implementation that serves both, and the geometry-free part is the whole persistence, graduation, pruning and classification state machine together with the periodicity test, not per-identifier periodicity modelling alone, which the FreMEn server already provides.
- *A Minimal Lifelong Classifier Assembled from Known Parts*: survival decay after the Persistence Filter (Rosen et al., 2016), a Schmitt-trigger graduation band motivated by Removert (Kim and Kim, 2020), negative-evidence ray clearing as in ReFusion (Palazzolo et al., 2019), and a parallel FreMEn-lite periodicity test at a fixed period (Krajník et al., 2017), emitting a four-class label from one SLAM-free module, with every equation transcribed from the shipped code and the implementation-versus-specification differences stated explicitly (§3.7).
- *A Calibrated Periodic Label*: a known moment-generating-function bound for quadratic forms of sub-Gaussian vectors (Hsu et al., 2012, Remark 2), applied to the floating-mean periodogram (Zechmeister and Kürster, 2009), bounds the probability that the periodic label fires on iid clutter under occupancy-independent sampling, for any noise rate, at one read-out (Proposition 3); spending the level over the touch count, a Bonferroni-type allocation in the spirit of alpha spending (Lan and DeMets, 1983), extends the bound to every read-out of an unpruned history (Proposition 4). The inequalities are standard; the contribution is their use for per-cell occupancy periodicity and the measurement that repeated read-outs inflate false Periodic labels without such a correction.
- *A Mechanistic, Reproducible Synthetic Characterisation*: calibration runs on dedicated calibration seeds (E0, and the calibration half of E5), disjoint from all evaluation seeds, four seeded experiments (E1–E4) that isolate backend equivalence, periodicity detection, the contribution of hysteresis and decay, and per-dimension cost, and a multi-seed, trajectory-level evaluation of the live detector with a four-class confusion (E5), with every table and figure regenerated from committed CSVs and failure modes diagnosed rather than averaged away.

Across the suite, both backends graduate a 161-cell wall by the third window and keep recall at 1.0 through window 39; their only quality gap, a final static precision of 0.9253 (`grid2d`) against 0.9758 (`voxel3d`) under 100 movers per window, arises in the backends' native sampling rather than in the shared classifier. We report and correct two defects of the first release's periodicity test, with before-and-after results. An uncentred Fourier amplitude could demote constant walls. An uncalibrated amplitude threshold labelled Bernoulli clutter Periodic at 0 to 4 of 5 cells depending on read-out length, a mean false-positive rate of 0.331 even after centring. Because the map re-tests every window, a constant significance level still let up to 953 of 2000 clutter cells be labelled Periodic at some window of a 1024-window run; with the level spent over the touch count, at most 2 of 2000 are, on held-out seeds, and the E2 false-positive rate is 0 at every read-out length from 8 to 100 windows. Both 50%-duty doors are still detected, for a true-positive rate of 2/3, but only

from 25 windows instead of 8, and the one miss is a provable interaction between pruning and the amplitude gate. Over 20 held-out scenes the dominant labelling error is a false Static label on aperiodic cells, not a false Periodic one: with the shipped defaults 1960 of 2000 Bernoulli(0.5) cells end Static, and the graduation rule cannot separate such clutter from a noisy wall of similar occupancy (Remark 1). Removing the hysteresis band raises flicker from 54 to 574 toggles at the same decay, and the estimated footprint is a flat 56 B per cell with periodicity off; `grid2d` is cheaper per call than `voxel3d` at every tested extent in every timing recording, by a factor that we do not report because it was not measured reliably, alongside 4.4–10.0× more live 3D cells. On this synthetic suite, the results are consistent with a lifelong-mapping core that is written once and shared across dimensions, with the two backends differing in how many cells their geometry creates. Whether the calibrated test improves on FreMEn's own component selection, and how the classifier behaves on real LiDAR data, remain to be evaluated (§5).

2. Related Work

Lifelong and persistent mapping. ELite (Gil et al., 2025) scores each point's ephemerality at two timescales and maintains a lifelong/static/delta map triad; LT-mapper (Kim and Kim, 2022) composes live, meta and delta maps through explicit alignment, removal and change-management stages. Khronos (Schmid et al., 2024) splits an active window from a long-term reconstruction with semantics, and Yang et al. (2024) keep versioned deltas over a base map so that past sessions can be rebuilt. Classifying cells by their temporal behaviour predates these systems. Temporal occupancy grids (Arbuckle et al., 2002) classify cells by occupancy over several timescales with planar laser range-finders and locate doors in a real-world setting; Mitsou and Tzafestas (2007) index every cell's occupancy history over time and separate static, low-dynamic and high-dynamic cells, in simulation with known poses. Multi-timescale and static/temporary map splits follow in Biber and Duckett (2005) and Meyer-Delius et al. (2010). Pomerleau et al. (2014) infer online whether each 3D point is static or dynamic from repeated observations, over seven months of data, and Berrio et al. (2022) describe a long-term map maintenance pipeline for autonomous vehicles. For 2D occupancy grids, Stefanini et al. (2022, 2023) update a LiDAR occupancy map for long-term operation while accounting for localisation error, with the aim of keeping the map current rather than modelling periodic change. Two 2026 works, concurrent with this one, produce layered outputs: MTD-Map (Kim et al., 2026) encodes the direction and duration of occupancy transitions per voxel in one stage and outputs static, dynamic and transition maps, evaluated on real datasets against established baselines, and Chen and Sun (2026) assign map content to long-term static, potentially dynamic and removed observed-dynamic layers from geometric and semantic evidence. Neither describes an explicit periodic class. STRATA implements only the graduation half of this pattern—one per-cell state machine with no scan matching, loop closure, session alignment, semantics or object layer—and is intended as the core such systems wrap, not a competitor; correspondingly, it lacks their real-data and field-scale evidence.

Persistence, forgetting and periodicity. The Persistence Filter (Rosen et al., 2016) estimates each feature's survival probability from a Bayesian survival-time model, and the Markov-chain occupancy models of Tipaldi et al. (2013) and Saarinen et al. (2012) learn per-cell transition rates. Perpetua (Saavedra-Ruiz et al., 2025) combines persistence and emergence filters under multiple hypotheses for features that disappear and reappear. FreMEn (Krajník et al., 2017), descending from spectral analysis of long-term maps (Krajník et al., 2014a), models cyclic occupancy with a few Fourier components selected by amplitude from a set of candidate periods, and it is the closest prior work to STRATA. Krajník et al. (2016) use it in a 2D spatio-temporal occupancy grid in which each cell represents both persistence (through the mean time between state changes) and periodicity; the grid integrates one

map per patrol, replaces the ROS map server, and predicts time-specific maps for localisation and planning. FROctomap (Krajník et al., 2014b) applies the FreMEn spectral model to 3D octree voxels, the exploration work of Krajník et al. (2015) displays the static and daily-periodic cells of such a grid, and the open-source FreMEn repository (STRANDS Project, 2017) provides separate 2D-grid and octree packages, and also a generic server that maintains FreMEn models of arbitrary binary states keyed by an identifier. Warped hypertime (Krajník et al., 2019) extends the idea to pseudo-periodic variation. STRATA is narrower than this family in period modelling: it tests one configured base period and its harmonics and does not select periods. It differs from the family in two respects: its primary output is a discrete class, reached through a hysteresis band and pruning, rather than a predicted occupancy probability; and it attaches a finite-sample false-alarm bound to the periodic label. It also updates once per window, whereas the grid of Krajník et al. (2016) integrates one map per patrol. STRATA uses the log-odds update of Thrun et al. (2005), clamped as in OctoMap (Hornung et al., 2013), replaces the survival posterior with one global multiplicative decay (a constant-rate special case), and runs a bounded FreMEn-style accumulator in parallel with graduation so that an oscillating cell is labelled Periodic instead of being mis-graduated or mis-pruned.

Dynamic-object removal and negative evidence. Removert (Kim and Kim, 2020) conservatively retains definite static points and then reverts wrongly removed ones; ERASOR (Lim et al., 2021) removes them offline from pseudo-occupancy ratios; ReFusion (Palazzolo et al., 2019) treats a reliably observed empty voxel as evidence. STRATA applies the same logic online and per cell: a graduated cell demotes only after sustained contradicting evidence (a hysteresis band rather than a revert pass), and free-space misses along each ray are the evidence that demotes it.

Map data structures and ROS 2 tooling. OctoMap (Hornung et al., 2013) and UFOMap (Duberg and Jensfelt, 2020) share the clamped log-odds core but compress space with octrees, the latter with an explicit unknown state; the voxel hash of Nießner et al. (2013) is the structure `voxel3d` follows. Nav2's spatio-temporal voxel layer (Macenski et al., 2020) decays costmap evidence for planning, layered costmaps (Lu et al., 2014) composite per-cell layers, and SLAM Toolbox (Macenski and Jambrecic, 2021) is a 2D SLAM library that includes prototype lifelong mapping. For swappable backends, ROS 2 offers `pluginlib` (Open Robotics, 2024); STRATA deliberately uses one interface and a two-way factory instead. Relative to the works above, the structural difference is narrow. Periodicity modelling that does not depend on geometry already exists, since the FreMEn server models any binary state that has an identifier (STRANDS Project, 2017). What STRATA shares, unchanged, between its 2D and 3D ray-casting backends is the full cell-level state machine: windowed log-odds with survival decay, graduation and demotion hysteresis, pruning, layer classification and the periodicity test.

Periodicity tests and error control. The periodic test uses the generalised Lomb–Scargle periodogram (Zechmeister and Kürster, 2009). Analytic false-alarm probabilities for periodogram peaks, including the search over frequencies, are derived under white Gaussian noise (Baluev, 2008). For binary series, Schmidtke and Vetter (2026) test a constant success probability against a periodic one of unspecified period, with a level that holds asymptotically. STRATA addresses a narrower question: a finite-sample, distribution-free bound for independent Bernoulli occupancy at one configured period and its harmonics, under any sampling pattern fixed independently of the occupancy. Proposition 3 obtains it from the moment-generating-function bound for quadratic forms of sub-Gaussian vectors of Hsu et al. (2012, Remark 2), followed by Markov's inequality and a union bound, with the variance proxy of Hoeffding's lemma (Hoeffding, 1963); the result is an application of that bound, not a new inequality. Spending the level over the touch count (Proposition 4) is a Bonferroni-type allocation over repeated looks, related to the alpha-spending functions of group-sequential trials (Lan and DeMets, 1983); time-uniform bounds from nonnegative supermartingales (Howard et al., 2020) are the modern alternative for repeated testing, which we do not use.

3. STRATA: A Geometry-Free Engine Beneath Two Map Geometries

STRATA is delivered as two ROS 2 packages with a hard dependency boundary. `strata_core` holds the entire occupancy, persistence and periodicity logic together with both geometry backends, in C++17 with Eigen as its only third-party dependency. `strata` is a thin node that owns message conversion, TF lookups, publishers, a save service, file I/O and input diagnostics (Tables 13 and 14). This section specifies the engine exactly as implemented: every equation is transcribed from `strata_core`, and §3.7 lists where the code departs from the design document. Table 17 collects the symbols.

Problem Setup: One Classifier, Any Geometry. A sensor sweep at integration tick k delivers an origin $\mathbf{o}_k \in \mathbb{R}^3$ and a set of hit endpoints $\mathcal{P}_k \subset \mathbb{R}^3$, both already in the map frame (the pose comes from an external localizer). A backend $b \in \{2D, 3D\}$ supplies two operators only: a key map $\kappa_b : \mathbb{R}^3 \to C \cup \{\perp\}$ onto integer cell ids $C \subset \mathbb{Z}$, and a ray walk $\rho_b(\mathbf{o}, \mathbf{p}) \subset C$ listing the free cells between origin and endpoint. Ticks are grouped into windows $w = 1, 2, \ldots$. Each live cell i carries a state $\mathbf{x}_i = (\ell_i, N_i, g_i, \boldsymbol{\phi}_i)$—log-odds occupancy, touched-window count, graduated flag and Fourier accumulators—and the map reads out a class $c_i \in \{\mathsf{U}, \mathsf{T}, \mathsf{P}, \mathsf{S}\}$ (Unknown, Transient, Periodic, Static). The design objective is that the per-window map update factor as

$$\mathcal{X}_w = \Phi\big(\mathcal{X}_{w-1},\ \Gamma_b(\{(\mathbf{o}_k, \mathcal{P}_k)\}_{k \in w})\big), \qquad c_i = \Psi(\mathbf{x}_i), \tag{1}$$

where only the evidence extractor Γ_b depends on the backend, while the engine Φ and the read-out Ψ are written once and see nothing but integer ids. The classes should make walls *durable* (S persists under clutter), movers *plastic* (T cells are forgotten), and cyclic structure *periodic* (P is neither baked into S nor pruned). The stages below instantiate Γ_b, Φ and Ψ; Algorithm 1 gives the complete window update.

3.1. Keying Observations: Reducing Geometry to Integer Cell Ids

For every endpoint $\mathbf{p} \in \mathcal{P}_k$ that the backend accepts ($\kappa_b(\mathbf{p}) \neq \perp$) it registers one miss on each cell of $\rho_b(\mathbf{o}_k, \mathbf{p})$ and then one hit on $\kappa_b(\mathbf{p})$; an endpoint it does not accept contributes nothing, not even its ray. The engine keeps only per-window counters,

$$h_{i,w} = \sum_{k \in w} \sum_{\mathbf{p} \in \mathcal{P}_k} \big[\kappa_b(\mathbf{p}) = i\big], \qquad m_{i,w} = \sum_{k \in w} \sum_{\mathbf{p} \in \mathcal{P}_k} \big|\{j \in \rho_b(\mathbf{o}_k, \mathbf{p}) : j = i\}\big|, \tag{2}$$

creating a cell lazily on first touch with $\ell_i = 0$, $N_i = 0$, $g_i = \text{false}$. This is Γ_b; it is the only backend-specific code.

Fixed 2D grid. `Grid2DBackend` addresses an array of W columns and R rows with resolution r and origin (x_0, y_0), drops z, and maps a point to a row-major index,

$$\kappa_{2D}(\mathbf{p}) = \left\lfloor \tfrac{p_y - y_0}{r} \right\rfloor W + \left\lfloor \tfrac{p_x - x_0}{r} \right\rfloor, \tag{3}$$

returning $\perp$ outside the array. An endpoint outside the array is dropped together with its ray: a beam that ends beyond the array registers no hit and clears no cell, including the cells it crosses inside the array. ρ_{2D} is integer Bresenham (Bresenham, 1965) from the sensor cell up to but excluding the hit cell, so the sensor cell is included and the hit cell receives only its hit. A sweep whose origin lies outside the array registers its hits but clears no ray. The node adds a filter of its own before the backend: its scan adapter drops every beam whose range is not finite or lies outside [`range_min`, `range_max`], so a beam without a return clears nothing, and a reading equal to `range_max` is kept as a hit.

Sparse 3D hash. `Voxel3DBackend` floor-divides each axis by the voxel size v, offsets it by 2^{20} to keep it non-negative, and packs the three 21-bit fields into one 64-bit key,

$$\kappa_{3D}(\mathbf{p}) = (u_x \ll 42) \,|\, (u_y \ll 21) \,|\, u_z, \qquad u_a = \lfloor p_a / v \rfloor + 2^{20}. \tag{4}$$

The map grows through the hash and needs no preallocated extent. Its key space is nevertheless finite: each 21-bit field represents the indices -2^{20} to $2^{20} - 1$ around the origin, far beyond the extent of a building at the default voxel size, and the code does not check this range, so a point outside it would receive a wrong key rather than being dropped. ρ_{3D} is a sample march rather than an exact voxel traversal: with $\mathbf{d} = \mathbf{p} - \mathbf{o}$ and $S = \lfloor \|\mathbf{d}\| / (v/2) \rfloor$ half-voxel steps, the interior samples $\mathbf{o} + s\,\mathbf{d}/\max(S, 1)$, $s = 1, \ldots, S-1$, are hashed with (4). This is simpler than exact traversal but can visit a voxel more than once along a long ray and can skip a thin one when S rounds down.

3.2. Accumulating Evidence: Windowed Log-Odds with Survival Decay

A window closes on integration tick k when $L \le 1$ or $k \bmod L = 0$, with L the `layer_interval`. The ROS node can instead close windows every `window_period_s` seconds of message time (`window_mode time`, the shipped setting; the code default and every experiment use L): windows are counted from the first header stamp, a message stamped past the open window first closes it and every silent window since, and a message with a zero stamp is dropped. A window that observes nothing changes no cell, so a silent gap of k windows is exactly k updates done in one pass; silent windows advance the phase without a touch, so within one uninterrupted stream of message time, and as long as message loss and TF drops are independent of occupancy, the false-alarm bound of §3.4 holds in either mode and only the period, and with it the power, depends on the window rule. A stamp more than half a window behind the newest one (a bag loop, a clock reset) or more than 10^6 windows ahead (an unset stamp followed by a real one) closes the open window and re-anchors the window clock, while cell histories carry on; neither the bound nor the persistence evidence holds across such a jump, so restart the node rather than loop a bag. Scan-counted windows last L divided by the rate of messages actually integrated (§5). The update then scores every live cell once, so a single hit outweighs any number of misses in the same window:

$$\tau_i = [h_{i,w} > 0 \vee m_{i,w} > 0], \qquad o_i = [h_{i,w} > 0], \qquad f_i = [h_{i,w} = 0 \wedge m_{i,w} > 0]. \tag{5}$$

For *touched* cells only ($\tau_i = 1$), one log-odds increment is applied, multiplied by the survival decay λ, and clamped:

$$\ell_i \leftarrow \min\Big(l_{\max}, \max\big(l_{\min},\ \lambda\,(\ell_i + o_i\, l_{\text{hit}} + f_i\, l_{\text{miss}})\big)\Big), \qquad N_i \leftarrow N_i + 1, \tag{6}$$

with occupancy probability $p_i = \sigma(\ell_i) = 1/(1 + e^{-\ell_i})$. The update is the inverse-sensor-model rule of Thrun et al. (2005) with a constant-rate special case of the Persistence Filter's forgetting (Rosen et al., 2016). The order matters: the decay multiplies the *already incremented* value, so the fresh evidence is itself attenuated in the same window.

Proposition 1 (Saturation under repeated hits)**.** *If a cell is hit in every window,* $0 < \lambda < 1$, $l_{\text{hit}} > 0$ *and* $l_{\min} \le 0 < l_{\max}$, *then* ℓ_i *converges monotonically from* 0 *to* $\min(l_{\max}, \ell^\star)$ *with* $\ell^\star = \lambda l_{\text{hit}}/(1 - \lambda)$.

Proof. Without the clamp the recursion is $\ell \mapsto \lambda(\ell + l_{\text{hit}})$, an affine contraction with factor λ and unique fixed point $\ell^\star$; from $\ell = 0 < \ell^\star$ the iterates increase monotonically towards it. The clamp is monotone and, since every iterate is non-negative and $l_{\min} \le 0$, only the upper bound can bind, so the clamped iterates increase to $\min(l_{\max}, \ell^\star)$. □

With the shipped defaults (Table 11; λ=0.97, l_{hit}=0.85) $\ell^\star \approx 27.5$, so the clamp binds at $l_{\max}$=5 and $p_{\max} = \sigma(5) \approx 0.9933$. The default thresholds p_{grad}=0.8 and p_{prune}=0.05 correspond to $\ell \geq \ln 4 \approx 1.386$ and $\ell < \ln(1/19) \approx -2.944$.

3.3. Gating Graduation: A Schmitt Trigger Against Flicker

After the log-odds step, p_i and the periodicity amplitude a_i (§3.4) are recomputed for *every* cell, touched or not, and the graduated flag follows a two-sided Schmitt trigger with two periodicity guards, evaluated in this order:

$$\text{graduate:} \quad \neg g_i \wedge \neg \pi_i \wedge p_i \geq p_{\text{grad}} \wedge N_i \geq N_{\min} \;\Rightarrow\; g_i \leftarrow \text{true}, \tag{7}$$

$$\text{demote:} \quad g_i \wedge \left(p_i \leq p_{\text{dem}} \vee \pi_i\right) \;\Rightarrow\; g_i \leftarrow \text{false}, \tag{8}$$

where π_i is the periodic predicate of (13). The interval $[p_{\text{dem}}, p_{\text{grad}}]$ is the hysteresis band: through the occupancy disjunct $p_i \leq p_{\text{dem}}$, a Static cell demotes only after sustained contradicting evidence, not after one contradicting window—the failure that Removert's revert pass repairs after the fact (Kim and Kim, 2020) is prevented online. The guard $\neg\pi_i$ keeps a periodic cell out of the static map; the term $\vee\, \pi_i$ removes a Static cell that later reveals periodicity. That term lies outside the band and has no hysteresis of its own. It cannot, however, fire without contradicting evidence: with the centred amplitude of §3.4, π_i requires a cell to have been observed free in a substantial share of its touched windows (Remark 2), so a wall that is occupied whenever it is seen is never demoted by it. The predicate also requires the noise-calibrated significance of Proposition 3, with its level spent over the touch count. A graduated cell is never pruned, so by Proposition 4 a wall whose occupancy is merely iid noise is demoted by this term with probability at most δ for as long as its history is not interrupted; a noisy wall that the occupancy disjunct demotes can afterwards be pruned and re-created, and the new history has a budget of its own. The term still acts in a single window when it does fire; E3 measures its cost in flicker with periodicity on (§4.4).

Remark 1 (Which occupancy rates graduate). The graduation rule sees a cell only through its log-odds. Take a cell that is touched in every window and occupied independently with probability m. While the clamp is inactive, (6) reads $\ell \leftarrow \lambda(\ell + x)$ with $\mathbb{E}\, x = m\, l_{\text{hit}} + (1 - m)\, l_{\text{miss}}$. For $\lambda < 1$ the mean of ℓ converges to $\lambda\, \mathbb{E}\, x/(1 - \lambda)$, which lies above the graduation threshold logit p_{grad} exactly when

$$m > m^\dagger = \frac{\frac{1-\lambda}{\lambda} \operatorname{logit} p_{\text{grad}} - l_{\text{miss}}}{l_{\text{hit}} - l_{\text{miss}}}. \tag{9}$$

For $\lambda = 1$ the same expression is the rate $-l_{\text{miss}}/(l_{\text{hit}} - l_{\text{miss}})$ above which the walk drifts upward. The shipped defaults give $m^\dagger = 0.354$ and the E2 parameters $m^\dagger = 0.5$. This is a statement about a mean, not a bound: it ignores the clamp and pruning, and fluctuations graduate cells below $m^\dagger$; E6 measures the transition (Table 9). Its consequence is that Static means no more than *occupied in more than roughly $m^\dagger$ of the touched windows*. The engine holds no evidence that separates a wall seen in 60% of its windows from clutter present in 50% of them.

Proposition 2 (No same-window oscillation). *If $p_{\text{dem}} < p_{\text{grad}}$, a cell that graduates by (7) in a window is not demoted by (8) in the same window.*

Proof. Graduation requires $\neg\pi_i$ and $p_i \geq p_{\text{grad}} > p_{\text{dem}}$, and p_i, π_i are not recomputed between (7) and (8), so both disjuncts of the demotion condition are false. □

3.4. Detecting Periodicity: A FreMEn-Lite Amplitude Test

In parallel with occupancy, every touched cell feeds an incremental Fourier model following FreMEn (Krajník et al., 2017). The model is restricted to one configured base period and its first harmonics; unlike FreMEn, it does not select periods from a candidate set, so a cycle whose frequency is not a harmonic of the base frequency is outside its scope. With window index w, base period T (in windows), H harmonics, $\omega = 2\pi/\max(1,T)$, occupancy sample $o_i \in \{0,1\}$ and $\theta_k = (k{+}1)\omega w$, the accumulators $\boldsymbol{\phi}_i = (n_i, S_{0,i}, C_{k,i}, S_{k,i}, E^c_{k,i}, E^s_{k,i})$ evolve as

$$n_i \mathrel{+}= 1, \quad S_{0,i} \mathrel{+}= o_i, \quad C_{k,i} \mathrel{+}= o_i \cos\theta_k, \quad S_{k,i} \mathrel{+}= o_i \sin\theta_k, \quad E^c_{k,i} \mathrel{+}= \cos\theta_k, \quad E^s_{k,i} \mathrel{+}= \sin\theta_k, \tag{10}$$

for $k = 0, \ldots, H{-}1$. Here n_i counts *touched* windows, not elapsed ones, and E^c, E^s record the phases at which the cell was touched. With the touched-window mean $\bar{o}_i = S_{0,i}/n_i$, the *centred* coefficients are $\alpha_k = 2(C_{k,i} - \bar{o}_i E^c_{k,i})/n_i$ and $\beta_k = 2(S_{k,i} - \bar{o}_i E^s_{k,i})/n_i$, i.e. $\alpha_k + \mathrm{i}\beta_k = \frac{2}{n_i}\sum_{w\in\mathcal{W}_i}(o_{i,w} - \bar{o}_i)\, e^{\mathrm{i}\theta_k}$ over the touched windows $\mathcal{W}_i$, the mean-removed spectrum of FreMEn. The harmonic amplitudes and the dominant amplitude are

$$a_{i,k} = \sqrt{\alpha_k^2 + \beta_k^2}, \qquad a_i = \begin{cases} 0, & n_i < T, \\ \max_{0\le k<H} a_{i,k}, & n_i \ge T, \end{cases} \tag{11}$$

and the periodic predicate π_i, which adds a significance test to the amplitude threshold, is given in (13) below. The phase prediction $\hat{p}_i(w) = \mathrm{clip}_{[0,1]}\big(S_{0,i}/n_i + \sum_k [\alpha_k \cos((k{+}1)\omega w) + \beta_k \sin((k{+}1)\omega w)]\big)$ is exposed for downstream use. The gate $n_i \ge T$ withholds a harmonic estimate until a period's worth of touches exists; a sparsely observed cell needs T *touches*, which can span many more than T windows.

Remark 2 (Centring, and the defect it corrects). Centring bounds the amplitude by the occupancy variation. Since $|o_{i,w} - \bar{o}_i|$ averages $2\bar{o}_i(1-\bar{o}_i)$ over $\mathcal{W}_i$, the triangle inequality gives

$$a_i \;\le\; 4\,\bar{o}_i(1-\bar{o}_i)$$

under any sampling of $\mathcal{W}_i$. A cell that is occupied (or free) in every touched window therefore has $a_i = 0$, and π_i with $a_{\min}$=0.3 requires $0.081 < \bar{o}_i < 0.919$: at least about 8% of its touched windows must be free. The code as first released (v0.1.0, commit `a82f935`; the package version string has not been raised since) omitted E^c, E^s, so its coefficients were uncentred. For an always-occupied cell, a_i was then twice the mean phase vector of $\mathcal{W}_i$: it measured *when* the cell was observed, not how its occupancy varied. For n consecutive touches this is

$$a_i = \max_{0\le k<H} \frac{2\left|\sin\left(n(k{+}1)\pi/T\right)\right|}{n\left|\sin\left((k{+}1)\pi/T\right)\right|} \;\le\; \frac{2}{n\sin(\pi/T)}.$$

It exceeds $a_{\min}$ at n = 10–13 for T=8 (peak 0.439 at n=11; the bound clears only from n=18), and at n = 29–40 for the shipped T=24, H=2 (bound: $n \ge 52$). Through the $\vee\,\pi_i$ term of (8), whose predicate was then $a_i \ge a_{\min}$ alone, a wall hit every window therefore left the Static layer for 120 integration ticks at L=10. Uneven phase coverage made this permanent: a wall seen in 8 of every 48 windows had $a_i \approx 1.66$. The corrected engine adds the two phase sums, 16 bytes per harmonic (the significance test below adds two more), and the per-cell cost order is unchanged. Ten regression tests cover both failures and genuine detection; seven of them fail on v0.1.0 and all pass after the fix (§4.1), and Table 3 compares E1–E3 before and after.

Calibrating the Periodic Test Against Noise. Centring removes the bias of the amplitude but not its variance, and $a_i \ge a_{\min}$ alone is not a test. If a cell's occupancy is pure noise, $o_{i,w} \sim \mathrm{Bernoulli}(m)$

independently of the phase, each centred coefficient has variance $2m(1-m)/n_i$ under uniform phase coverage, a standard deviation of $\sqrt{0.5/8} = 0.25$ at $m{=}\frac{1}{2}$, $n_i{=}T{=}8$. A fixed $a_{\min}$ is then crossed by chance at short lengths, and pruning (§3.5) keeps returning cells to short lengths. We therefore test each harmonic for significance. Write $c_w = \cos\theta_k$, $s_w = \sin\theta_k$ over the touched windows, $\tilde{\mathbf{z}}_w = (c_w - \bar{c},\ s_w - \bar{s})^\top$ for the centred phase vectors, $\mathbf{y}_{i,k} = \sum_{w\in\mathcal{W}_i}(o_{i,w} - \bar{o}_i)\,\tilde{\mathbf{z}}_w = \frac{n_i}{2}(\alpha_k, \beta_k)^\top$ and $\mathbf{M}_{i,k} = \sum_w \tilde{\mathbf{z}}_w\tilde{\mathbf{z}}_w^\top$. Then

$$d_{i,k} = \mathbf{y}_{i,k}^\top \mathbf{M}_{i,k}^{-1}\mathbf{y}_{i,k}, \qquad B(r) = \begin{cases} 2r\,e^{1-2r}, & r > \frac{1}{2}, \\ 1, & r \le \frac{1}{2}, \end{cases} \tag{12}$$

where $d_{i,k}$ is the drop in squared residual when a sinusoid at harmonic k is fitted with a floating mean, the generalised Lomb–Scargle periodogram (Zechmeister and Kürster, 2009). $\mathbf{M}_{i,k}$ needs two more accumulators per harmonic, $\sum\cos 2\theta_k$ and $\sum\sin 2\theta_k$, since $\sum c_w^2 = \frac{1}{2}(n_i + \sum\cos 2\theta_k)$ and $\sum c_w s_w = \frac{1}{2}\sum\sin 2\theta_k$. The periodic predicate requires, for some harmonic, both an effect size and significance at a nominal level δ (`periodic_false_alarm`):

$$\pi_i = \texttt{enable_periodicity} \wedge n_i \ge T \wedge \exists k:\ \big[a_{i,k} \ge a_{\min}\ \wedge\ H\,B(d_{i,k}) \le \delta\big]. \tag{13}$$

Proposition 3 bounds the false-alarm probability of this predicate. It is an application of known results, not a new inequality: the moment-generating-function step is the bound of Hsu et al. (2012, Remark 2) for a quadratic form of a sub-Gaussian vector, here with the identity matrix and Hoeffding's variance proxy $\frac{1}{4}$, and the rest is Markov's inequality and a union bound over the harmonics. We state it with the explicit constants that the shipped test uses.

Proposition 3 (Calibrated false-alarm bound)**.** *Let the touched windows $\mathcal{W}_i$ ($|\mathcal{W}_i| = n$) be fixed independently of the occupancy, and let $o_{i,w}$, $w \in \mathcal{W}_i$, be independent* Bernoulli(m) *for one unknown $m \in [0,1]$. If $\mathbf{M}_{i,k}$ is non-singular, then* $\Pr(d_{i,k} \ge r) \le B(r)$ *for every $r > 0$. Consequently, for every $a_{\min} \ge 0$ and $0 < \delta < H$, the predicate* (13) *is true with probability at most δ.*

Proof. Because $\sum_w \tilde{\mathbf{z}}_w = \mathbf{0}$, the unknown mean drops out: $\mathbf{y}_{i,k} = \sum_w \xi_w\tilde{\mathbf{z}}_w$ with $\xi_w = o_{i,w} - m$. Put $\mathbf{b}_w = \mathbf{M}_{i,k}^{-1/2}\tilde{\mathbf{z}}_w$ and $\mathbf{u} = \sum_w \xi_w\mathbf{b}_w$, so that $d_{i,k} = \|\mathbf{u}\|^2$ and $\sum_w \mathbf{b}_w\mathbf{b}_w^\top = \mathbf{I}_2$. Each ξ_w has mean zero and lies in an interval of length 1, so Hoeffding's lemma (Hoeffding, 1963) gives $\mathbb{E}\,e^{t\xi_w} \le e^{t^2/8}$, and by independence $\mathbb{E}\,e^{\mathbf{v}^\top\mathbf{u}} \le \exp\big(\frac{1}{8}\sum_w(\mathbf{v}^\top\mathbf{b}_w)^2\big) = e^{\|\mathbf{v}\|^2/8}$ for every $\mathbf{v}\in\mathbb{R}^2$. Following Hsu et al. (2012), let $\mathbf{g}\sim\mathcal{N}(\mathbf{0},\mathbf{I}_2)$ be independent of $\mathbf{u}$. For $0 \le \eta < 2$, $\mathbb{E}_{\mathbf{g}}\,e^{\sqrt{2\eta}\,\mathbf{g}^\top\mathbf{u}} = e^{\eta\|\mathbf{u}\|^2}$, so exchanging the expectations,

$$\mathbb{E}\,e^{\eta d_{i,k}} = \mathbb{E}_{\mathbf{g}}\,\mathbb{E}_{\mathbf{u}}\,e^{\sqrt{2\eta}\,\mathbf{g}^\top\mathbf{u}} \le \mathbb{E}_{\mathbf{g}}\,e^{\eta\|\mathbf{g}\|^2/4} = (1-\eta/2)^{-1}.$$

Markov's inequality gives $\Pr(d_{i,k}\ge r) \le (1-\eta/2)^{-1}e^{-\eta r}$, and $\eta = 2 - 1/r$ (admissible for $r > \frac{1}{2}$) yields $2r\,e^{1-2r}$; for $r \le \frac{1}{2}$ the bound 1 is trivial. B decreases strictly from 1 to 0 on $(\frac{1}{2},\infty)$, so $H\,B(d_{i,k}) \le \delta$ holds exactly when $d_{i,k} \ge r^\star$ with $B(r^\star) = \delta/H$. The union bound over the H harmonics then gives $\Pr(\pi_i) \le H\cdot\delta/H = \delta$. □

The bound needs no knowledge of m: the sub-Gaussian constant $\frac{1}{4}$ is the Bernoulli worst case $m = \frac{1}{2}$. When $\mathbf{M}_{i,k}$ is singular (for instance at a Nyquist harmonic), the code projects onto its range; the same argument with a one-dimensional $\mathbf{g}$ gives the smaller bound $(1-\eta/2)^{-1/2}e^{-\eta r}$, so B stays valid. Under uniform phase coverage at a harmonic with $2(k{+}1) \not\equiv 0 \pmod T$ (so that the sine column does not vanish; the defaults $T/H \in \{8/3, 24/2\}$ satisfy it) $\mathbf{M}_{i,k} = \frac{n}{2}\mathbf{I}_2$ and $d_{i,k} = n\,a_{i,k}^2/2$, so (13) becomes an amplitude threshold that shrinks with n,

$$a_{i,k} \ \ge\ \max\big(a_{\min},\ a^\star(n)\big), \qquad a^\star(n) = \sqrt{2r^\star/n}. \tag{14}$$

For δ=0.1 and H=3, $r^\star = 3.115$, so $a^\star(8) = 0.883$ and $a^\star(64) = 0.312$. The bound is conservative: in the Gaussian limit at $m{=}\frac{1}{2}$, $\Pr(d_{i,k} \geq r) \to e^{-2r}$, a factor $2e\,r$ below $B(r)$, or about 17 at $r^\star$. We therefore first treated δ as a nominal level for one read-out and chose it on a calibration set disjoint from every evaluation seed (E0, §4.1). The pre-stated rule took the largest $\delta \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ whose measured per-read-out false-alarm rate stays at or below 0.01, both over a null grid and in the live pipeline, and it selected δ=0.1 (Table 4). That rule compares point estimates: the worst pipeline rate at δ=0.1, 7 of 1000 cells, has a 95% Wilson upper limit of 0.0144 (Wilson, 1927), so the calibration does not establish the 0.01 target with confidence. Only δ is held out; the test itself was introduced after the amplitude-only rule produced false positives on the E2 seed, so E2 is not a blind evaluation of its design. More importantly, one read-out is not what the map does.

Spending the Level Over Repeated Read-Outs. The map evaluates (13) after every window, so a clutter cell gets one chance per read-out, and pruning (16) erases its history and starts another. E5 measures the consequence: at a single-read-out level of 0.1, up to 953 of 2000 Bernoulli clutter cells are labelled Periodic at some window within 1024 windows (§4.3). The shipped rule therefore replaces the constant level δ in (13) by a level that decreases with the touch count,

$$\delta_n = \delta\,\frac{T}{n(n+1)} \quad (n \geq T), \qquad \sum_{n \geq T} \delta_n = \delta, \tag{15}$$

where the sum telescopes because $T/(n(n{+}1)) = T/n - T/(n{+}1)$. This is a Bonferroni-type allocation of the level over the sequence of read-outs, in the spirit of the alpha-spending functions used for repeated significance tests in group-sequential trials (Lan and DeMets, 1983); Proposition 4 is the union bound over it.

Proposition 4 (Trajectory-level false-alarm bound)**.** *Call a* history *of cell i its touched windows from its creation until it is pruned or the run ends. Suppose that, given everything observed before the history begins, its touched windows are fixed independently of its occupancy and its occupancy samples are independent* Bernoulli(m). *With the level δ_n of* (15) *in* (13)*, the probability that π_i holds at some read-out of the history is at most δ, whatever the length of the run. If L_N histories of the cell begin in windows $1, \ldots, N$, the probability that the cell is labelled Periodic at some window up to N is at most $\delta\,\mathbb{E}[L_N]$.*

Proof. Condition on everything observed before the history begins; under the hypothesis the history is then a fixed-design sample. For $n \geq T$ let D_n be the event that the statistic of its first n samples, computed as if the cell were never pruned, satisfies (13) at level δ_n. Proposition 3 with δ_n in place of δ gives $\Pr(D_n) \leq \delta_n$. A read-out at touch count n evaluates π_i on exactly these n samples (read-outs between two touches repeat the last value), and it happens only if the history survived to n, so the event that π_i ever holds is contained in $\bigcup_{n \geq T} D_n$. The union bound gives at most $\sum_{n \geq T} \delta_n = \delta$; pruning only removes read-outs. For the second claim, the ℓ-th history begins at a window determined by the past, so the probability that it begins by N and fires is at most δ times the probability that it begins by N; summing over ℓ gives $\delta\,\mathbb{E}[L_N]$. □

The proposition is strongest where pruning does not act. A graduated cell is exempt from (16), so while a wall stays Static its history is not interrupted, and the probability that the $\vee\,\pi_i$ term of (8) ever demotes it on iid noise during that history is at most δ. A noisy wall can still be demoted through the occupancy disjunct, pruned and re-created (Bernoulli(0.6) cells begin 5 histories in 1024 windows under the E2 parameters, Table 18), so over a whole run the bound for a wall is $\delta\,\mathbb{E}[L_N]$ as well. For clutter that is pruned and re-created, the bound grows with the number of histories, which depends on the data: a Bernoulli(0.4) cell under the E2 parameters begins about 62 histories in 1024 windows, and $\delta\,\mathbb{E}[L_N]$ is then vacuous. That regime is measured (E5), not guaranteed. The null is still that of

Proposition 3: temporally correlated clutter violates it, and E5 shows that the test flags such clutter. The shipped level is chosen on separate calibration seeds, now at trajectory level. The pre-stated rule takes the largest $\delta \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ for which the rate of ever labelling a Bernoulli clutter cell Periodic within 1024 windows has a Wilson 95% upper limit (Wilson, 1927) of at most 0.01 in every calibration configuration. It selects δ=0.2, the largest candidate (Table 6), and this is the shipped default. For the shipped rule, then, what is proved is the per-history bound 0.2; the 0.01 target is a measurement. Under uniform coverage the threshold of (14) becomes $a^\star(n) = \sqrt{2r_n^\star/n}$ with $B(r_n^\star) = \delta_n/H$, which falls roughly like $\sqrt{\log n/n}$ rather than $1/\sqrt{n}$. The detection delay is the price: in E2 a 50%-duty door with T=8 is detected from 25 windows, against 15 with a single-read-out level of 0.1 and 8 without the test.

3.5. Classifying and Pruning: Four Classes from One State

At the end of the same per-cell pass, a cell with little evidence is erased together with its Fourier accumulators:

$$\text{erase } i \iff \neg g_i \wedge p_i < p_{\text{prune}} \wedge a_i < a_{\min}. \tag{16}$$

Static cells and periodic cells are therefore never pruned. The guard uses the lenient effect-size screen $a_i < a_{\min}$ rather than $\neg\pi_i$, so a *candidate* periodic cell keeps its accumulators, and hence its growing n_i, until the test resolves it. Every other erased cell loses $\boldsymbol{\phi}_i$. We considered keeping the Fourier history of erased cells and rejected it: pruning exists so that clutter does not accumulate, and a record for every cell ever touched would grow without bound, fastest in `voxel3d`, where every sampled free voxel is a cell. A re-created cell restarts at $n_i = 0$ with a new history and, under (15), a new level budget; this is why the trajectory bound of Proposition 4 scales with the number of histories. The read-out Ψ of (1) is a priority ladder,

$$c_i = \begin{cases} \mathsf{U} & i \text{ absent}, \\ \mathsf{S} & g_i, \\ \mathsf{P} & \pi_i, \\ \mathsf{T} & p_i \geq p_{\text{prune}}, \\ \mathsf{U} & \text{otherwise}, \end{cases} \tag{17}$$

evaluated top to bottom. The node renders it with the convention Unknown −1, free 0, Transient 50, Periodic 75, Static 100, writing in ascending priority so Static wins any tie: `grid2d` keeps one byte per cell outside the classifier, an observed-free bit set by the ray clear and never cleared, a bit for a hit in the open window and a bit for whether the cell's last observation was a hit; it draws 50 for a cell hit in the open window or last observed as a hit, or a Transient cell whose evidence leans occupied ($p_i > 0.5$), and 0 for a cleared cell last observed free, so free means last observed free; `voxel3d` publishes only Static voxel centres. Table 15 lists every transition. The update visits every live cell, including untouched ones, because (7)–(16) read the recomputed p_i and a_i; its cost per window is linear in the number of live cells times H, and between window closes a sweep costs $O(1)$ per endpoint plus the ray walk. Per-cell state is a 24-byte record plus roughly 32 bytes of hash-node overhead, the ~56 B/cell used in §4. That figure is an estimate for the periodicity-off path, not a measurement. A Fourier record, with two scalar accumulators and six heap-allocated vectors of H values in a second hash map, is allocated only for cells touched with periodicity enabled and is not included in it, nor are the hash tables' bucket arrays or the `grid2d` flag byte per grid cell with its list of cells touched in the open window.

Algorithm 1: `endWindow`: one layered update per closed window, as in `layered_map.cpp`

Input: live cells $\mathcal{X}$ with window counters h_i, m_i; window index w

$w \leftarrow w + 1$;
foreach *live cell* i **do**
  $\tau \leftarrow [h_i{>}0 \vee m_i{>}0]$, $o \leftarrow [h_i{>}0]$, $f \leftarrow [h_i{=}0 \wedge m_i{>}0]$; // Eq. (5)
  if τ **then**
    $\ell_i \leftarrow \mathrm{clamp}\big(\lambda(\ell_i + o\, l_{\mathrm{hit}} + f\, l_{\mathrm{miss}}),\, l_{\min},\, l_{\max}\big)$, $N_i \leftarrow N_i + 1$; // Eq. (6)
    if `enable_periodicity` **then** update $\boldsymbol{\phi}_i$ with (o, w) // Eq. (10);
  $p_i \leftarrow \sigma(\ell_i)$, $a_i \leftarrow \mathrm{amplitude}(\boldsymbol{\phi}_i)$, $\pi_i \leftarrow$ Eq. (13) on $\boldsymbol{\phi}_i$; // Eqs. (11)-(13)
  if $\neg g_i \wedge \neg\pi_i \wedge p_i \geq p_{\mathrm{grad}} \wedge N_i \geq N_{\min}$ **then** $g_i \leftarrow$ true // Eq. (7);
  if $g_i \wedge (p_i \leq p_{\mathrm{dem}} \vee \pi_i)$ **then** $g_i \leftarrow$ false // Eq. (8);
  $h_i \leftarrow 0$, $m_i \leftarrow 0$;
  if $\neg g_i \wedge p_i < p_{\mathrm{prune}} \wedge a_i < a_{\min}$ **then** erase i and $\boldsymbol{\phi}_i$ // Eq. (16);

Output: updated $\mathcal{X}$; classes by Eq. (17) on demand

3.6. Selecting a Backend: One Interface, One String

`MapBackend` exposes five methods: `integrate(obs, origin)` runs Γ_b for one sweep, `tick()` advances the integration counter and may close a window, `closeWindows(k)` closes k windows at once for the time-based rule of §3.2, and two read-only counters report Static and Transient cells. The node reads one string parameter, `backend` $\in$ {`grid2d`, `voxel3d`}, holds one `unique_ptr` per concrete backend and instantiates exactly one at construction (the string is not validated: any value other than `grid2d` selects `voxel3d`); all engine parameters are read through one shared helper, so the tuning surface is backend-independent by construction. At two backends a runtime `dlopen` registry such as `pluginlib` (Open Robotics, 2024) would add discovery machinery whose cost is not yet repaid. Because Φ and Ψ in (1) never see coordinates, any behavioural or cost difference between the backends must originate in κ_b, ρ_b or their input sampling—the property the experiments test.

3.7. Reconciling Code and Specification

We document the code that ships. Two places diverge from the design document (`SPEC.md`), and each changes operational semantics.

Remark 3 (Forgetting is coupled to re-observation). The design document states that every cell decays every window. In the code, (6) runs only for touched cells, so a cell outside the sensor's field of view neither decays nor relaxes toward the prior: its ℓ_i freezes. Clutter fades only while it is re-observed as free, and anything that leaves the field of view persists (§5).

Remark 4 (The amplitude gate counts touches). The design document gates the amplitude on *elapsed* windows; the code gates on the touched-window count n_i of (13). This is the mechanism behind the low-duty-cycle miss in §4.2.

4. Experiments

We ask five questions. Do the two backends behave as one engine wearing two geometries (E1, E4)? Does the shared classifier detect periodic structure, and how does it fail (E2)? Does the live, pruned and repeatedly tested detector control false alarms over a whole run (E5)? Which cells does the

graduation rule call Static when they are neither walls nor periodic (E5, E6)? Which mechanism actually stabilises the static layer (E3)?

4.1. Experimental Setup

Harness and protocol. All experiments are driven by a standalone harness that compiles directly against the `strata_core` sources, with no ROS 2, ament or colcon in the loop; it lives in `paper/experiments/`. One command, `bashpaper/experiments/run_all.sh`, builds the core and its unit tests, checks the `ctest` suite (one target aggregating the 56 gtest cases: ten regression tests for the amplitude defect of Remark 2, nine for the significance test of Proposition 3 and eight for the spent level of Proposition 4), builds and runs the calibration E0 and the experiments E1–E6, and writes the CSVs in `paper/experiments/results/`; every table and plot in this paper is regenerated from those files (Appendix A). E0–E3, E5 and E6 reproduce their committed files byte for byte with GCC and libstdc++, and E1, one E0 file and E6's confusion change under another standard library (Appendix A); E4's timings are wall-clock and are recorded separately. Every stochastic experiment E1–E4 derives its `std::mt19937` generator from the constant seed 12345 plus a per-run offset recorded in code (offsets 0–1500), and the seed is logged in each CSV. The calibration E0 uses seeds from 20260928 upward, E5 draws its calibration seeds from 30260928 and its evaluation seeds from 40260928 upward, so no calibration stream coincides with an evaluation stream; E6 reuses the scene seeds of E5's confusion and draws its sweep from 50260928 upward, and the unit tests use a further disjoint range. E1–E4 magnitudes are single-seed point estimates: we report no cross-seed dispersion for them, and gaps such as 0.9253 against 0.9758 illustrate a mechanism rather than a distributional estimate. E5 and E6 are multi-seed and report Wilson 95% intervals.

What is compared. There is no external baseline. The lifelong-SLAM systems of §2 do not expose a persistence core that this harness could drive. The FreMEn per-cell model (Krajník et al., 2016, 2017; STRANDS Project, 2017), with its amplitude-ranked component selection, could be driven by the same streams, but we have not run it, so every comparison below is internal. The centred amplitude-only rule kept in Table 3 thresholds FreMEn-style coefficients at one fixed period; it is STRATA's own earlier rule, not an implementation of FreMEn. Instead E1 and E4 compare the two STRATA backends, each driven in its native geometry by its own seeded stream (`grid2d` on the integer lattice, `voxel3d` continuously); because both call the identical engine, any divergence arises in κ_b, ρ_b or their input sampling. E2 and E3 drive `LayeredMap` directly with no backend at all, which by (1) characterises the classifier for whichever backend supplies the ids. Each harness sets its own parameters in code (Table 12 for E1–E4; E0, E5 and E6 use the E2 parameters or the shipped defaults, as stated with each); in particular every experiment, E0 and E5 included, runs with one window per tick (L=1), so *window* and *frame* coincide.

Metrics. Static-set precision, recall and F1 against a known wall set (E1, E3); per-read-out false-alarm rates on pure-noise cells and the read-out length from which a door stays Periodic (E0); the trajectory-level false-alarm rate, i.e. the share of non-periodic cells labelled Periodic at *some* window up to a horizon N, the per-window labelling rate, and four-class precision and recall (E5); Periodic-class true- and false-positive rates against ground truth and the dominant amplitude a (E2); *flicker*, the number of per-window toggles of a wall cell's Static flag (E3); the share of Bernoulli(m) cells labelled Static (E6); final live-cell count, estimated memory at 56 B per cell and, where it was recorded reliably, latency per `integrate()` call (E4).

Scenarios and hardware. E0 runs first and fixes the single-read-out level (§3.4): iid Bernoulli(m) streams at $T/H \in \{8/3, 24/2\}$, $m \in \{0.05, 0.1, 0.2, \ldots, 0.9, 0.95\}$ and 12 read-out lengths $n \in \{8, \ldots, 100\}$ for T=8 and 8 lengths $n \in \{24, \ldots, 240\}$ for T=24 (listed in `e0_calibration.cpp`),

with 20,000 streams per point; 3000 Bernoulli(m) clutter cells ($m \in \{0.3, 0.5, 0.7\}$) through the live `LayeredMap` with the E2 parameters, read out at every n = 8–100; and four doors (4-on/4-off, 2-on/2-off, 3-on/5-off, and 4-on/4-off with 10% flipped windows) in 40 runs each at random phase offsets. E1: a 161-cell static cross (the segments y=60 and x=60 over [20, 100]) hit every window, plus 5, 25 or 100 fresh random movers per window, for 40 windows, on a 120×120 grid and a voxel map, both at resolution 1.0, once with periodicity off and once with it on at the shipped T=24, H=2. E2: three periodic doors (period 8 at 4-on/4-off and 2-on/6-off, period 4 at 2-on/2-off), a constant wall and four Bernoulli(0.5) aperiodic controls, with T=8, H=3, read out at 64 windows and, in a sweep, after every length n = 8–100 (a fresh map per length). E3: 50 wall cells observed occupied with probability 0.6 per window among 10 movers per window, 80 windows, the same noise replayed for all 36 configurations of $p_{\text{grad}} \in \{0.6, 0.7, 0.8, 0.9\} \times p_{\text{dem}} \in \{0.3, 0.5, p_{\text{grad}}\} \times \lambda \in \{0.90, 0.97, 1.0\}$, with periodicity off and again with it on at the shipped defaults. E5: Bernoulli(m) clutter, $m = 0.1, \ldots, 0.9$, plus always-occupied, always-free and temporally correlated cells (a two-state Markov chain with mean 0.5 and mean dwell 10 windows), 2000 cells per configuration (10 seeds of 200), through the live `LayeredMap` with the E2 and with the shipped parameters, touched every window or independently with probability q=0.5, over 1024 windows, labelled after every window; four door types for detection delay; and 20 E2-style scenes for the confusion of Table 8. The calibration half of E5 fixes the shipped level on its own seeds before the evaluation half runs. E6, added after E5 showed many false Static labels, repeats that confusion on the same 20 scene seeds under the E2 parameters with λ=0.97 and under the shipped defaults (whose doors have the same duty cycles at three times the period, since T=24), and sweeps Bernoulli(m) cells over $m = 0.05, 0.10, \ldots, 0.95$, 2000 cells per point (10 seeds of 200), touched every window for 256 windows, under four parameter sets: E2, E2 with λ=0.97, the shipped defaults, and the shipped defaults with $l_{\text{miss}} = -l_{\text{hit}}$. All E1–E4 results are from the current engine (centred coefficients and the test at δ=0.2 spent over the touch count). The same harness run against three earlier engines is kept for comparison (Table 3): v0.1.0, the centred engine with the amplitude-only rule $a \geq a_{\min}$, and the test at the single-read-out level δ=0.1. E4: 150 random hits per window for 20 windows at nominal extents 100^2, 250^2 and 500^2, resolution 1.0, periodicity off, each configuration run 15 times on a fresh map after one discarded warm-up run. E4 alone reads the wall clock. Timings are to be taken on a quiet host with `paper/experiments/measure_e4.sh`: the script refuses to start unless the 1-minute load average is below 0.50, pins the benchmark to one core, discards one warm-up run per configuration, records the load and the largest CPU consumers before and after, and marks a recording invalid when the load afterwards is 1.50 or more. No such recording exists, so we report no latency and no ratio. Its live-cell counts follow only from ray geometry and are deterministic.

4.2. Main Results

Backend equivalence of the static layer (E1). With $N_{\min}$=3 a cell needs three touched windows before it may graduate, and recall jumps from 0 to 1.0 exactly at the third window (t=2) for all six backend–clutter combinations, then holds at 1.0 through t=39: clutter never demotes a wall cell (Table 1, Figure 2). Table 1 is the periodicity-off run. With periodicity on at the shipped defaults (T=24, H=2, δ=0.2 spent over the touch count) the run is identical window for window: the wall is occupied whenever it is touched, so its centred amplitude is 0 (Remark 2) and $d_{i,k} = 0$. The v0.1.0 engine instead lost the whole wall from the Static layer in windows 29–40 (t = 28–39), in all six runs, and ended with Static recall 0 (Table 3). The only errors are false Static cells created when a random mover re-hits one cell often enough to graduate on its own. At 5 movers per window this never happens and precision stays at 1.0; at 25 it saturates at one spurious cell (precision 0.9938); at 100 false statics keep accumulating, from the first extra cell at t=17 (`grid2d`) or t=25 (`voxel3d`) to 174 or 165 predicted cells at t=39. This yields the one material gap between backends, precision 0.9253

Table 1: Static-layer quality (E1) against the 161-cell ground-truth wall set. *Recall 1 from* is the first window index t at which recall reaches 1.0 (it never drops afterwards); precision, F1 and the predicted static-set size are at the final window t=39. Single seed (12345), so no dispersion is reported. Bold marks the better backend under high clutter.

Backend	Clutter (movers/window)	Recall 1 from	Precision	F1	$\lvert\hat{\mathcal{S}}\rvert$
`grid2d`	low (5)	t=2	1.0000	1.0000	161
`grid2d`	med (25)	t=2	0.9938	0.9969	162
`grid2d`	high (100)	t=2	0.9253	0.9612	174
`voxel3d`	low (5)	t=2	1.0000	1.0000	161
`voxel3d`	med (25)	t=2	0.9938	0.9969	162
`voxel3d`	high (100)	t=2	**0.9758**	**0.9877**	165

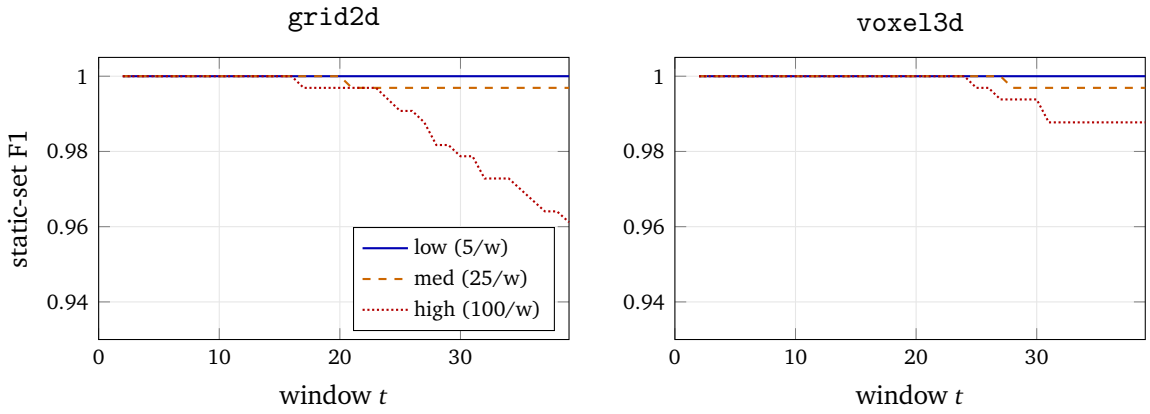


Figure 2: Static-layer F1 over time (E1). F1 is 0 for $t < 2$ (no cell has met $N_{\min}$=3; not shown) and reaches 1.0 at t=2 for every backend and clutter level. Only the 100-movers-per-window regime accumulates false statics, more so on `grid2d`. Plotted directly from `e1_static_quality.csv`.

against 0.9758. Both runs apply the identical graduation rule, each to its own natively sampled mover stream (movers on the integer lattice for `grid2d`, continuous for `voxel3d`, both confined to one z-plane), so the gap reflects quantisation and sampling rather than the classifier; we do not claim it isolates a single geometric mechanism.

Single-read-out calibration (E0). Table 4 applies the single-read-out rule of §3.4 (level constant in n). The Chernoff bound is conservative at every candidate level. At δ=0.1 the worst null rate over all grid points is 0.0050 and the worst pipeline rate 0.007, both within the 0.01 target, whereas δ=0.2 exceeds it (0.0149 and 0.014). The selected single-read-out level is therefore δ=0.1; the shipped rule replaced it (E5). Relaxing δ from 0.01 to 0.1 shortens the median read-out length at which a 4-on/4-off door becomes stably Periodic from 22 to 15 windows. The Gaussian-residual approximation of the periodogram, the textbook false-alarm formula (Zechmeister and Kürster, 2009), fails the same check: at a level of 0.01 it reaches a null rate of 0.037 for n near T, where binary sequences put real mass on exactly periodic patterns. No E1–E4 seed was used in E0.

Periodicity detection (E2). Both 50%-duty doors are classified Periodic, with reference amplitudes 0.653 and 0.707 and Bonferroni-adjusted bounds below 10^{-3} (Table 2). The constant wall is Static with amplitude exactly 0 at every length. No aperiodic control is Periodic, so at the n=64 read-out the Periodic class reaches a true-positive rate of 2/3 and a false-positive rate of 0/5; the two earliest engines scored 1/5 (v0.1.0) and 2/5 (centred only) at this length. Figure 4 and Table 3 show that this is not a favourable read-out. Over all lengths n = 8–100 the pipeline has no false positive at any

Table 2: Periodicity detection (E2) after 64 windows with the shipped test, $a_{\min}$=0.3, δ=0.2 spent over the touch count, T=8, H=3. *Ref. amplitude* and *Ref. HB* (the Bonferroni-adjusted Chernoff bound of Proposition 3; Periodic needs $\leq \delta_n$) come from a non-pruning reference model fed the identical occupancy stream. Periodic TPR = 2/3, FPR = 0/5. *Outcome* scores the Periodic class; a non-periodic cell ending Static is not a Periodic false positive but is a wrong map label (*false Static*).

Probe cell	Ground truth	Final class	Ref. amplitude	Ref. HB	Outcome
`door_p8_4on4off` (50% duty)	periodic	Periodic	0.653	$< 10^{-3}$	correct
`door_p8_2on6off` (25% duty)	periodic	Transient	0.462	$< 10^{-3}$	**miss**
`door_p4_2on2off` (50% duty)	periodic	Periodic	0.707	$< 10^{-3}$	correct
`wall_constant`	non-periodic	Static	≈ 0	1	correct
`aperiodic_0`	non-periodic	Static	0.159	1	**false Static**
`aperiodic_1`	non-periodic	Unknown	0.329	0.056	correct
`aperiodic_2`	non-periodic	Transient	0.139	1	correct
`aperiodic_3`	non-periodic	Transient	0.251	0.585	correct

length, as with the single-read-out level, against a mean false-positive rate of 0.363 (v0.1.0) and 0.331 (centred), with 0–4 of 5 per length in both. The gain is paid for in detection delay. The true-positive rate is 0 for n = 8–19 and 1/3 at n = 20–24, and it reaches 2/3 at n = 25; the single-read-out level reached it at n = 15 and the two earliest engines from n=8. The mean true-positive rate over the sweep falls from 0.667 to 0.624 (single read-out) and 0.563 (spent).

Two further observations qualify the zero. First, it is no longer owed to pruning. With the single-read-out level, a non-pruning reference model fed the same streams and read with the same predicate called `aperiodic_1` Periodic at 25 of 93 lengths (n = 29–31, 45–50 and 52–67), a mean false-positive rate of 0.054: at n=64 this cell's amplitude is 0.329 and its bound $HB = 0.056 < 0.1$, a genuine tail event of this seed, roughly 3.7 null standard deviations. With the spent level the same reference model (dotted in Figure 4) has no false positive at any length, because $\delta_{64} = 0.2 \cdot 8/(64 \cdot 65) \approx 3.8 \times 10^{-4}$. Second, `aperiodic_0` ends Static. With λ=1 its log-odds is a random walk that reached p_{grad}, and it was previously masked by a spurious Periodic label (centred engine) that demoted it. E2 does not score the Static class; E5 does, over 20 scenes, and finds this failure common (Table 8). The amplitude curves confirm the gate of (11): for the 4-on/4-off door the amplitude is exactly 0 at n=6 < T, 0.653 at every sampled multiple of T ($n = 8, 16, \ldots, 64$) and 0.581 at n=12, where the uncentred v0.1.0 amplitude had overshot to 0.871 (Figure 3). The miss, the 25%-duty door, is not a failure of the test: the reference model detects it at every length from n=57 with the spent level (from n=26 at a single read-out), with amplitude 0.462 at n=64. The live pipeline never gets there, for a reason that follows from the update rule.

Proposition 5 (Prune/maturity race)**.** *Under the E2 parameters (l_{hit}=1, l_{miss}=−1, λ=1, p_{prune}=0.05, T=8), a cell touched every window with the 2-on/6-off pattern is erased by (16) within at most 7 consecutive touched windows of its creation. Hence $n_i < T$ throughout every lifetime, $a_i = 0$, and the cell is never classified Periodic.*

Proof. With λ=1 and unit increments, ℓ_i is an integer walk from 0 (the clamp at ±5 never binds below). Since $\sigma(-2) \approx 0.119 \geq p_{\text{prune}} > \sigma(-3) \approx 0.047$, erasure happens exactly when ℓ_i first reaches −3, provided $a_i < a_{\min}$, which holds while $n_i < T$. If the cell is created with $j \geq 3$ vacant windows left in its cycle, it reaches −3 after 3 windows. If $j \leq 2$ vacant windows remain, ℓ_i reaches $-j$, then $2 - j$ after the two occupied windows, then −3 after $5 - j$ further vacant windows: 7 windows in total. If it is created in an occupied window, $\ell_i \leq 2$ after at most two hits and −3 after five more misses, again at most 7. In every case the lifetime is at most 7 < T=8 touched windows, so $a_i = 0$ by (11) and π_i is false. Erasure also needs $\neg g_i$, which holds because $\ell_i \leq 2$ throughout, so $p_i \leq \sigma(2) \approx 0.881 < p_{\text{grad}}$=0.9 and

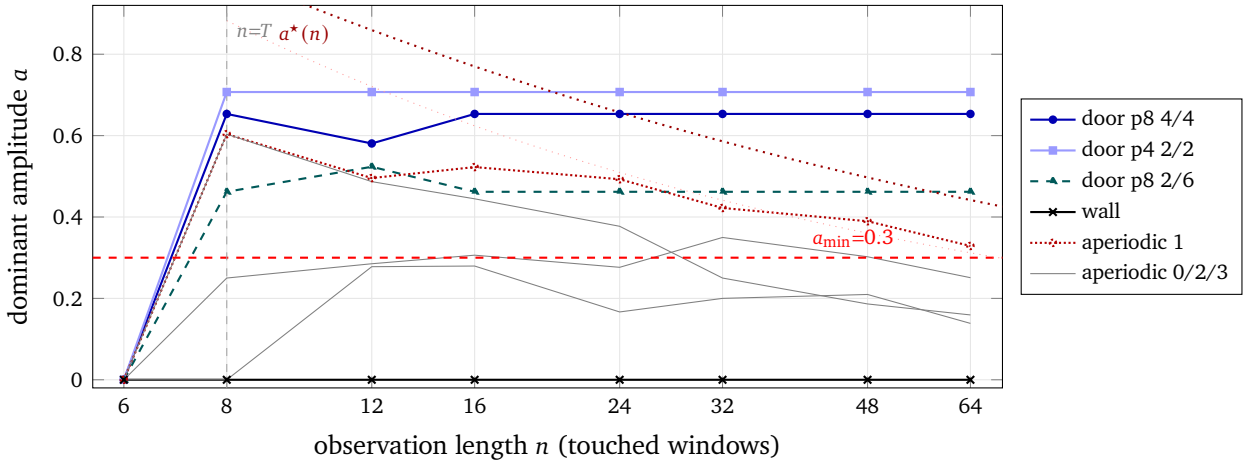


Figure 3: FreMEn-lite amplitude against observation length (E2), from a non-pruning reference model of the corrected engine fed each probe cell's occupancy stream (T=8, H=3). The $n \geq T$ gate forces a=0 at n=6; both 50%-duty doors settle well above $a_{\min}$, and the 25%-duty door would too, yet the live pipeline prunes it first (Proposition 5). The constant wall is exactly 0 at every length because the coefficients are centred (Remark 2); the Bernoulli controls hover near $a_{\min}$. The thick dotted curve $a^\star(n)$ is the amplitude the shipped test requires under uniform phase coverage, with δ=0.2 spent over the touch count and H=3 (Eq. (14) with δ_n of Proposition 4); the thin dotted curve is the single-read-out threshold at δ=0.1. A cell is Periodic only above both $a_{\min}$ and $a^\star(n)$. Plotted from `e2_amplitude_vs_length.csv`.

the cell never graduates. Erasure discards $\boldsymbol{\phi}_i$, so each re-created cell starts again from $n_i = 0$. □

The race couples the prune rule (16) to the touched-window gate; the reference model, which never prunes, is not subject to it.

Per-dimension cost of the shared engine (E4). In all 13 timing files that exist, `grid2d`'s `integrate()` is cheaper than `voxel3d`'s at every tested extent. All of them were taken on a loaded host or marked invalid, and in those the ratio of median latencies varies between files (9.9–27.6× at 100^2), so only the direction is reported; all files are kept in `results/e4_exploratory/` and `results/e4_recordings/` (Table 5, Figure 5). The live-cell counts are exact: `voxel3d` ends each run holding 10.0×, 6.5× and 4.4× as many cells, because its half-voxel ray march creates a hash entry for every traversed free voxel, whereas exact 2D Bresenham clears a bounded strip. We measured neither ray steps nor the per-cell update cost under each backend separately, so we do not attribute the latency difference to one cause. Memory is not measured: the harness multiplies the live-cell count by a fixed 56 B per cell (per-cell evidence and hash node with periodicity off; table and backend overhead excluded), which gives 37.0 MB for the largest 3D configuration (661,410 cells).

4.3. Trajectory-Level False Alarms and Confusion (E5)

What a deployed map does. E0 and E2 read a cell once per run. The map labels every cell after every window, so the event that matters for a deployed map is that a non-periodic cell is labelled Periodic at *some* window of the run. E5 measures this event directly on held-out seeds, for the live detector with pruning, re-creation and re-testing, over horizons of 64, 256 and 1024 windows (Table 7; every configuration in Table 18).

The single-read-out level does not control it. With δ=0.1 applied at every read-out, up to 84 of 2000 Bernoulli clutter cells are labelled Periodic within 64 windows and 953 of 2000 within 1024 (E2 parameters, m=0.4, 95% interval [0.4547, 0.4984]). The per-window labelling rate stays small,

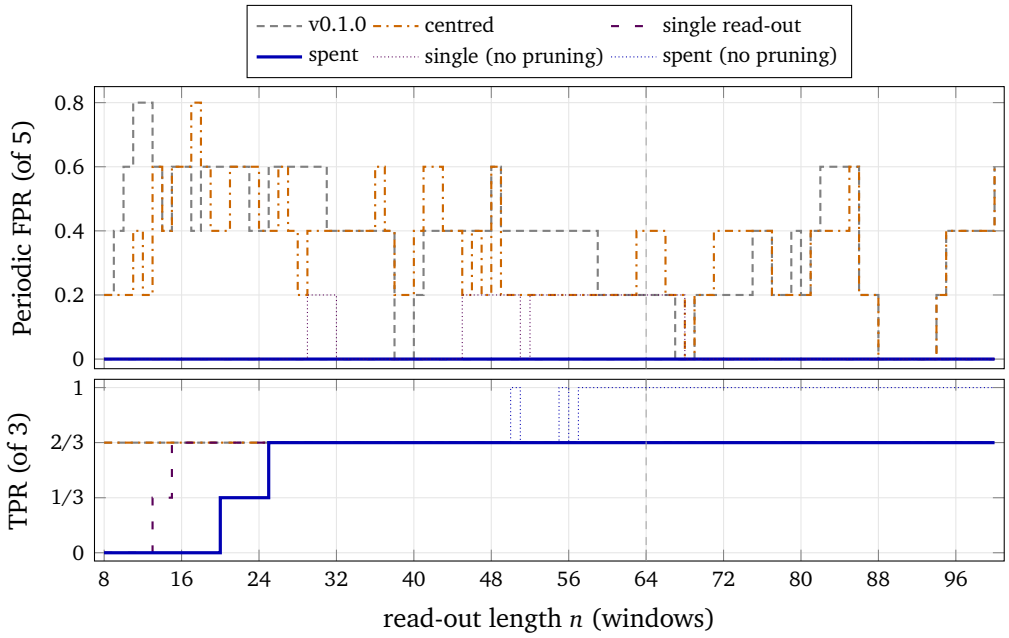


Figure 4: Periodic false- and true-positive rates against read-out length (E2), from the real `LayeredMap`, a fresh map per length, T=8, H=3, $a_{\min}$=0.3, for the four engine versions of Table 3. The dotted lines apply the single-read-out and the spent predicate to a non-pruning reference model fed the same streams (lower panel: spent only). The dashed vertical marks the n=64 read-out of Table 2. The v0.1.0 and centred TPR curves coincide at 2/3. Plotted from `e2_rates_vs_length.csv` and its three archived counterparts.

at most 0.0047, but it accumulates over the run. The shipped parameters are affected less, because their slower log-odds walk prunes Bernoulli clutter less often, yet 202 of 2000 cells still flip at some window (m=0.3).

The spent level does. With δ=0.2 spent over the touch count, chosen on the calibration seeds (Table 6), at most 2 of 2000 Bernoulli clutter cells are ever labelled Periodic within 1024 windows in every held-out configuration, a Wilson 95% upper limit of 0.0036, and the per-window rate is at most 3.9×10^{-6}. Always-occupied and always-free cells are never labelled Periodic under either rule. The proved bound of Proposition 4 covers the never-pruned case at 0.2. For pruned clutter the proved bound $\delta\,\mathbb{E}[L_N]$ is vacuous (a Bernoulli(0.4) cell under the E2 parameters begins about 62 histories), and the measured rate sits far below it: the Chernoff bound is conservative, and successive read-outs of one history are strongly correlated, so the union bound over them is loose.

Correlated clutter is flagged. The Markov cells violate the independence the bound assumes: their occupancy has real power at low frequencies. Under the shipped T=24 the spent test labels 1908 of 2000 of them Periodic within 1024 windows (1998 at a single read-out); at T=8 it labels 2. The periodogram test cannot tell a slow random process from a period of 24 windows; this is a limitation of the null, not a failure of the bound.

What it costs. On held-out seeds the median window from which a noiseless 4-on/4-off door stays Periodic moves from 15 to 25, a 2-on/2-off door from 13 to 20, a 3-on/5-off door from 21 to 31, and a 4-on/4-off door with 10% flipped windows from 26 to 48; at the shipped T=24 a 12-on/12-off door stays at 24 windows, and with 10% flips it moves from 25 to 40. No door run fails to become Periodic.

Four-class confusion. Table 8 scores all four labels over 20 held-out scenes. Under the E2 parameters walls are Static and dynamic cells Transient or Unknown, with one exception in 1000. No cell other than a door is Periodic. All 800 50%-duty doors are Periodic and none of the 400 25%-duty doors,

Table 3: Effect of the three corrections to the periodicity test, same harness, seeds and parameters. *v0.1.0*: uncentred amplitude and the amplitude-only rule $a \geq a_{\min}$ (results in `results/pre_fix_2026-09-28/`); *Centred*: centred amplitude, amplitude-only rule (`results/pre_calibration_2026-09-28/`); *Single*: the significance test of Proposition 3 at δ=0.1 at every read-out (`results/pre_spending_2026-09-28/`); *Spent*: the level δ=0.2 spent over the touch count (Proposition 4, current engine). E1 and E3 with periodicity on use the shipped T=24, H=2, $a_{\min}$=0.3; E2 uses T=8, H=3. E1 counts are identical in all six backend–clutter runs.

	Periodicity	Periodicity on			
Measure	off	v0.1.0	Centred	Single	Spent
E1: windows $t \geq 3$ with Static recall < 1	0 windows	12 (t=28–39)	0 windows	0 windows	0 windows
E1: Static recall at t=39	1	0	1	1	1
E2 at n=64: TPR / FPR	–	2/3 / 1/5	2/3 / 2/5	2/3 / 0/5	2/3 / 0/5
E2, n = 8–100: mean FPR	–	0.363	0.331	0.000	0.000
E2, n = 8–100: FP count range (of 5)	–	0–4	0–4	0–0	0–0
E2, n = 8–100: lengths with no FP	–	10 of 93	7 of 93	93 of 93	93 of 93
E2, n = 8–100: lengths with constant wall Periodic	–	4 of 93	0 of 93	0 of 93	0 of 93
E2, n = 8–100: mean TPR	–	0.667	0.667	0.624	0.563
E2: TPR 2/3 at every $n \geq$	–	8	8	15	25
E2, n = 8–100: lengths with FP, no pruning	–	–	–	25 of 93	0 of 93
E3: total flicker, 36 configurations	3672	7132	5472	3792	3672
E3: flicker at p_{grad}=0.9, p_{dem}=0.3, λ=0.90	54	138	102	58	54
E3: flicker at p_{grad}=p_{dem}=0.9, λ=0.90	574	582	596	576	574

Table 4: Calibration of the nominal level δ (E0) on seeds disjoint from E1–E4. *Null*: worst false-alarm rate over the grid of iid Bernoulli(m) streams ($T/H \in \{8/3, 24/2\}$, $m \in [0.05, 0.95]$, $n \geq T$, 20,000 streams per point); *Pipeline*: worst per-read-out rate of 3000 clutter cells in the live `LayeredMap` (E2 parameters, pruning on, n = 8–100); *Door*: median read-out length from which a 4-on/4-off door stays Periodic (40 phases). Rule, fixed before the run: the largest δ with both rates ≤ 0.01. For comparison, the Gaussian-residual approximation of the periodogram at level 0.01 reaches a null rate of 0.037.

δ	Null rate	Pipeline rate	Door, median first n	Selected
0.01	0.0004	0.001	22	no
0.02	0.0007	0.002	20	no
0.05	0.0022	0.003	17	no
0.1	**0.0050**	**0.007**	**15**	**yes**
0.2	0.0149	0.014	14	no (rate > 0.01)

which Proposition 5 proves are never Periodic; the recall of the Periodic class is therefore a property of the scene mix, and we report counts per cell type rather than a recall. Within 256 windows the single-read-out level labels 219 of 2000 aperiodic cells Periodic at some window, the spent level 1. The dominant error is elsewhere: 1221 of 2000 aperiodic Bernoulli(0.5) cells end falsely Static, a share of 0.611 [0.589, 0.632].

False Static labels (E6). The periodicity test does not decide this error; the graduation rule does, and decay does not remove it. With λ=0.97 added to the E2 parameters, 1042 of 2000 aperiodic cells still end Static. With the shipped defaults 1960 of 2000 do, a share of 0.980 [0.973, 0.985] (1983 after 256 windows), and so do 11 of 1000 dynamic cells; of the 1200 doors of that scene 1091 are Periodic after 64 windows and all 1200 after 256 (Table 8). With λ=0.97 all 1200 doors of the E2 scenes are Periodic, the 25%-duty ones included; we did not analyse why the race of Proposition 5 is not lost there. Remark 1 gives the reason for the Static labels: the shipped l_{hit}=0.85 and l_{miss}=−0.4 put the mean-field threshold at $m^{\dagger} = 0.354$, so a cell that is occupied in half of its windows drifts up to the clamp. Table 9 measures the transition. With the shipped defaults, 396 of 2000 Bernoulli(0.25)

Table 5: Size of the map each backend builds from the same number of hits (E4): 150 random hits per window, 20 windows, resolution 1.0, periodicity off. Live-cell counts and memory (at 56 B per cell) follow from the seeded geometry and reproduce exactly; the ratio is voxel3d over grid2d at equal extent. No latency is tabulated (see text).

Backend	Extent	Live cells	Memory
`grid2d`	100^2	8,553	479 KB
`voxel3d`	100^2	85,399	4.78 MB
ratio		*10.0*×	
`grid2d`	250^2	45,657	2.56 MB
`voxel3d`	250^2	295,300	16.5 MB
ratio		*6.5*×	
`grid2d`	500^2	150,096	8.41 MB
`voxel3d`	500^2	661,410	37.0 MB
ratio		*4.4*×	

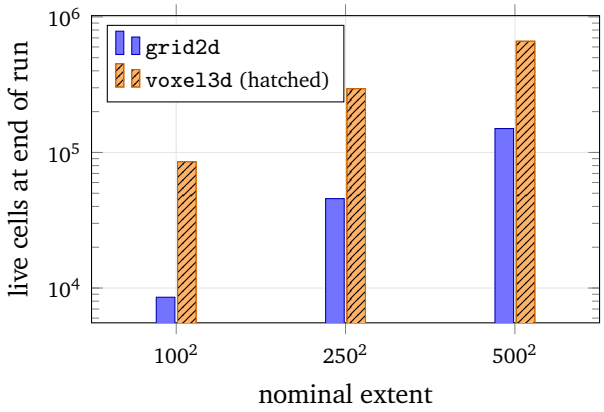


Figure 5: Live-cell count at the end of the run (E4), log scale. The counts are exact and follow from the seeded geometry. Plotted from `e4_throughput.csv`.

cells are Static after 256 windows, 783 at m=0.30, 1279 at 0.35, 1698 at 0.40 and 1974 at 0.50; and 485 of 2000 Bernoulli(0.10) cells are Static at some window. Symmetric increments ($l_{\text{miss}} = -l_{\text{hit}}$) move $m^\dagger$ to 0.525 and the count at m=0.50 to 1156, but they cost noisy walls: of Bernoulli(0.60) cells, which E3 scores as walls, 1741 of 2000 are then Static, against 1998 with the shipped defaults. No parameter set of the sweep separates the two populations. On the 0.05 grid the share of Static cells rises from below 10% to above 95% over an interval of m that is 0.30 wide with the shipped defaults and 0.35 wide with the other three sets. A user who knows the occupancy rate of the clutter to reject and of the walls to keep can place $m^\dagger$ between them through l_{miss} and λ, as far as the two rates lie further apart than that interval. We measured only these four sets, on iid cells touched in every window, and changed no default.

4.4. Ablation Studies

Hysteresis band versus decay (E3). E3 ablates the two stabilisers of the static layer on identical replayed noise (Table 10, full sweep in Table 16). Removing the hysteresis band has the largest effect in the sweep: at p_{grad}=p_{dem}=0.9 and λ=0.90, the worst configuration, the wall layer flickers 574 times and final F1 drops to 0.810 (recall 0.68), whereas the same noise and decay with the band widened to 0.6 (p_{dem}=0.3) flickers 54 times at F1 1.000, a more than tenfold reduction attributable to the

Table 6: Choice of the spent level δ (E5) on calibration seeds disjoint from every evaluation seed. *Worst ever-Periodic*: over 36 configurations (E2 and shipped parameters, touched every window or with probability 0.5, Bernoulli(m) clutter, $m = 0.1, \ldots, 0.9$, 2000 cells, 1024 windows), the largest count of cells labelled Periodic at some window, with the Wilson 95% upper limit of that rate. *Door*: median first window from which a 4-on/4-off door (E2 parameters, T=8) or a 12-on/12-off door (shipped, T=24) stays Periodic. Rule, fixed before the run: the largest δ whose upper limit is ≤ 0.01 in every configuration.

δ	Worst ever-Periodic (of 2000)	Wilson upper	Door, T=8 / T=24	Selected
0.01	0	0.0019	34 / 33	no
0.02	0	0.0019	32 / 30	no
0.05	0	0.0019	30 / 26	no
0.1	1	0.0028	27 / 24	no
0.2	**2**	**0.0036**	**25 / 24**	**yes**

Table 7: Trajectory-level false alarms of the live detector on held-out seeds (E5). Each entry is the worst Bernoulli(m) clutter configuration over $m = 0.1, \ldots, 0.9$: the number of 2000 cells (10 seeds of 200) labelled Periodic at some window up to N, with its Wilson 95% interval for N=1024, and the worst per-window labelling rate. q: probability that a cell is touched in a window, independent of its occupancy. *Single*: δ=0.1 at every read-out; *Spent*: δ=0.2 spent over the touch count. *Markov*: temporally correlated clutter (two-state chain, mean 0.5, mean dwell 10 windows), outside the null of Proposition 3. Constant cells were never labelled Periodic.

Parameters, touches	Rule	N=64	N=256	N=1024	95% interval, N=1024	Per window	Markov, N=1024
E2, every window	Single	84	280	953 (m=0.4)	[0.4547, 0.4984]	0.0045	349
	Spent	0	1	2 (m=0.4)	[0.0003, 0.0036]	3.9×10^{-6}	2
E2, q=0.5	Single	20	127	543 (m=0.4)	[0.2525, 0.2914]	0.0047	79
	Spent	0	0	2 (m=0.4)	[0.0003, 0.0036]	3.9×10^{-6}	0
Shipped, every window	Single	64	66	202 (m=0.3)	[0.0885, 0.1150]	0.0008	1998
	Spent	2	2	2 (m=0.4)	[0.0003, 0.0036]	1.0×10^{-6}	1908
Shipped, q=0.5	Single	18	52	102 (m=0.3)	[0.0422, 0.0615]	0.0009	1331
	Spent	1	1	1 (m=0.3)	[0.0001, 0.0028]	3.4×10^{-6}	409

band alone. Among zero-band configurations at λ=0.90, flicker rises with the coincident threshold (250 at 0.7, 343 at 0.8, 574 at 0.9), even though a higher threshold is usually the more conservative choice: a threshold pair with no gap is not a substitute for a band wherever it sits. Decay is second order. Holding the worst pair fixed, flicker falls as forgetting slows (574 at λ=0.90, 150 at 0.97, 122 at 1.0), because faster decay lets noisy log-odds cross the coincident threshold more often; with the widest band, decay barely matters (52–54 toggles across all three settings). Hysteresis, not decay, is what keeps the static layer stable (Figure 6).

Periodicity on (E3). Rerunning the sweep with periodicity on at the shipped defaults leaves final F1 unchanged in every configuration (last column of Table 16), and flicker is now identical to that with periodicity off in every configuration. The total over the 36 configurations is 3672 in both cases, against 3792 with the single-read-out level, 5472 with the centred amplitude-only rule and 7132 with v0.1.0. The widest-band configuration at λ=0.90 has 54 toggles, against 58 (single read-out), 102 (centred) and 138 (v0.1.0). These walls are free in 40% of their windows by construction, so their centred amplitude is sampling noise. Under the amplitude-only rule it crossed $a_{\min}$ often, and the unbanded $\vee\, \pi_i$ demotion (§3.3) toggled the wall out of the Static layer. With the single-read-out level, 1 or 2 such demotions remained per configuration, because each wall was re-tested in every window. With the spent level, the probability that π_i demotes a wall during one uninterrupted history is at most δ (Proposition 4); none is demoted in any configuration.

Table 8: Four-class confusion on 20 held-out scenes, touched every window, read out after 64 windows with the shipped test (E5 for the E2 parameters, E6 for the others; same seeds). Per scene: 50 walls, 20 doors of each of three types at random phase, 100 aperiodic Bernoulli(0.5) cells, and 50 dynamic cells crossed by a mover with probability 0.1 per window. Doors: 4-on/4-off, 2-on/6-off and 2-on/2-off for the E2 parameter sets (T=8); the same duty cycles at three times the period for the shipped defaults (T=24). *Single P*: Periodic count under the single-read-out rule and, after the slash, cells labelled Periodic at some window up to 256 under the single-read-out and the spent rule (E2 parameters only). The recall of a class over this fixed mix of cell types is a property of the mix, so counts are reported per type.

Parameters	Ground truth (cells)	Static	Periodic	Transient	Unknown	Single P / ever P (single, spent)
E2 (λ=1)	wall (1000)	1000	0	0	0	0 / 0, 0
	door (1200)	0	800	400	0	800 / 800, 800
	aperiodic (2000)	1221	0	672	107	12 / 219, 1
	dynamic (1000)	1	0	705	294	0 / 1, 0
aperiodic cells falsely Static: 0.611 [0.589, 0.632] (Wilson 95%); after 256 windows 1223 of 2000						
E2 with λ=0.97	wall (1000)	1000	0	0	0	–
	door (1200)	0	1200	0	0	–
	aperiodic (2000)	1042	0	861	97	–
	dynamic (1000)	0	0	686	314	–
aperiodic cells falsely Static: 0.521 [0.499, 0.543] (Wilson 95%); after 256 windows 1019 of 2000						
Shipped defaults	wall (1000)	1000	0	0	0	–
	door (1200)	0	1091	18	91	–
	aperiodic (2000)	1960	0	40	0	–
	dynamic (1000)	11	0	917	72	–
aperiodic cells falsely Static: 0.980 [0.973, 0.985] (Wilson 95%); after 256 windows 1983 of 2000						

5. Discussion and Limitations

What the evidence shows. Across E1–E4 the two backends behave as one engine wearing two geometries. E1's near-identical F1 trajectories and E4's live-cell counts are the empirical face of (1): `LayeredMap` and `PeriodicityModel`, exercised directly in E2 and E3, never see which backend produced the id they update. The backends diverge in two places, the high-clutter precision gap and the per-call cost. Both runs execute the same classifier code on differently sampled input; neither difference was isolated to one mechanism.

Cost Analysis. The window update is linear in the number of live cells (§3.5), so cost is governed by how many cells a backend creates. The flat voxel hash is the source of `voxel3d`'s 4.4–10.0× larger live-cell count: every sampled free voxel becomes its own entry, at an estimated 56 B with periodicity off, with no spatial compression. OctoMap (Hornung et al., 2013) and UFOMap (Duberg and Jensfelt, 2020) collapse coherent free or occupied regions into single octree nodes and would scale better to very large, sparsely occupied 3D extents. We accept the trade for $O(1)$ insert and lookup and a backend simple enough to unit-test without ROS; even the largest configuration tested needs 37.0 MB.

Forgetting outside the field of view. Remark 3 is an operational limitation, not a footnote. Decay runs only inside the touched branch, so a transient object that accumulates strong evidence and then leaves sensor coverage for good—a box dragged into a corridor the robot no longer visits—keeps its evidence indefinitely. Forgetting in STRATA is coupled to re-observation, not to elapsed time. In a bounded, frequently re-swept environment, the setting E1–E4 implicitly assume, this rarely matters; in a sparsely revisited one it is a real failure mode that the current code does not address.

Free space is remembered. The same coupling shows in `grid2d`'s free value. A cell that a ray has cleared reads free while its last observation was free and no Periodic, Static or occupied-leaning Transient class claims it, so free means last observed free, not currently free. An obstacle that appears

Table 9: Bernoulli(m) cells labelled Static (E6): of 2000 cells per entry (10 seeds of 200), touched every window, the number Static after 256 windows and, in parentheses, the number Static at some window up to 256. *Shipped, symmetric*: the shipped defaults with $l_{\text{miss}} = -l_{\text{hit}}$. $m^{\dagger}$ is the mean-field threshold of Remark 1. E3 scores Bernoulli(0.6) cells as walls, for which Static is the correct label; E5 scores Bernoulli(0.5) cells as clutter, for which it is an error.

m	E2 (λ=1)	E2, λ=0.97	Shipped	Shipped, symmetric
$m^{\dagger}$	0.500	0.534	0.354	0.525
0.10	0 (65)	0 (44)	11 (485)	7 (395)
0.20	32 (788)	21 (605)	172 (1767)	51 (1618)
0.25	65 (1356)	44 (1098)	396 (1973)	120 (1900)
0.30	138 (1735)	87 (1559)	783 (1995)	196 (1986)
0.35	288 (1921)	177 (1857)	1279 (2000)	320 (1998)
0.40	481 (1996)	367 (1975)	1698 (2000)	537 (2000)
0.50	1206 (2000)	1043 (2000)	1974 (2000)	1156 (2000)
0.60	1797 (2000)	1695 (2000)	1998 (2000)	1741 (2000)
0.70	1965 (2000)	1950 (2000)	2000 (2000)	1965 (2000)
0.80	1998 (2000)	1997 (2000)	2000 (2000)	1999 (2000)

Table 10: Hysteresis and decay ablation (E3), selected rows of the 36-configuration sweep on identical replayed noise (full sweep in Table 16). Band = $p_{\text{grad}} - p_{\text{dem}}$; flicker counts per-window `isStatic` toggles over 80 windows. Bold marks the lowest flicker in the sweep and the worst configuration.

p_{grad}	p_{dem}	Band	λ	Flicker ↓	F1 ↑
0.9	0.9 (deg.)	0.0	0.90	**574**	**0.810**
0.8	0.8 (deg.)	0.0	0.90	343	0.947
0.7	0.7 (deg.)	0.0	0.90	250	0.980
0.9	0.9 (deg.)	0.0	0.97	150	1.000
0.9	0.9 (deg.)	0.0	1.00	122	1.000
0.9	0.3	0.6	0.90	54	1.000
0.9	0.3	0.6	0.97	**52**	**1.000**
0.9	0.3	0.6	1.00	**52**	**1.000**

where the sensor no longer looks reads free until it is observed; from its first hit it reads 50, out of view too, until a ray observes the cell free again. Free space is also carved only along beams that return a hit inside the array (§3.1): a beam without a return, or one that ends beyond the array, clears nothing, so open space bounded only by such beams stays Unknown, and occupied evidence along them is never contradicted. A map smaller than the sensor's range therefore keeps cells that a model clearing every ray would release. The bit costs one byte per grid cell; `voxel3d` publishes no free space.

Scan-counted windows depend on the input rate. With `window_mode scans` a window is `layer_interval` integrated messages, so its length in seconds, and the period that `period_windows` stands for, follow the rate of messages the node actually integrates; messages lost on a best-effort link, dropped for missing TF or skipped by an overloaded node stretch every window. In the synthetic ROS check, withholding 40 % of the scans turns the tested door cells static (at least 95 %, none periodic) with scan windows (`grid2d`), while time windows keep the door periodic in `grid2d` and out of the static map in `voxel3d`; the continuous integration (CI) asserts all three. Time windows trade this for rate-dependent evidence: a window with few messages is less likely to see a hit, so a sensor rate that varies with the phase of a cycle can itself look periodic, outside the null of the test. Once a second the node publishes on `/diagnostics` the input rate, the lost share (estimated per sensor frame from

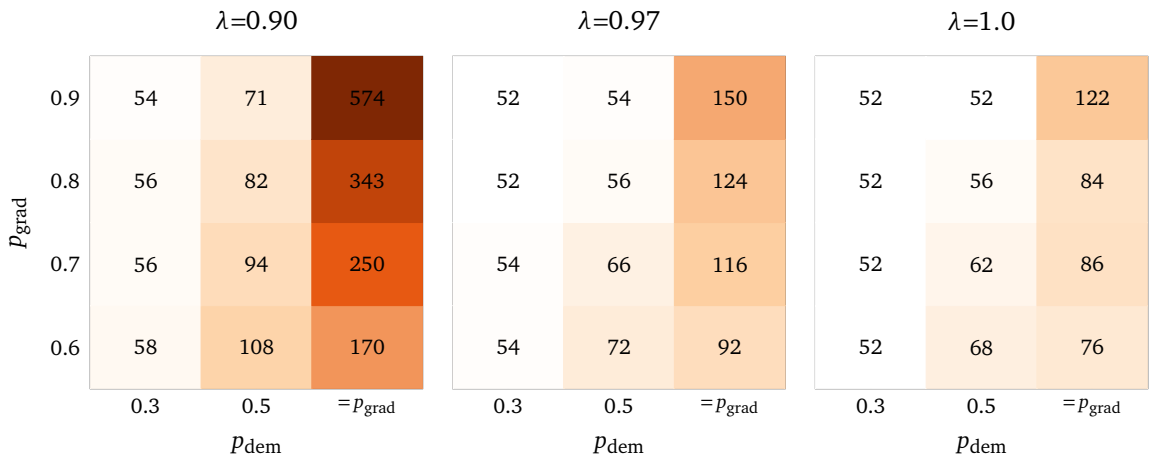


Figure 6: Flicker transitions across the E3 sweep (colour: $\log_{10}$ flicker; annotation: raw count). The zero-band column (p_{dem}=p_{grad}) holds the worst cell of every panel; widening the band suppresses flicker far more than slowing decay does. Plotted from `e3_sensitivity.csv`.

stamp gaps, valid while about a fifth of consecutive messages arrive back to back; above about 80 % loss, set `expected_scan_rate_hz`), TF failures, the window duration and the period in seconds. It warns on lossy or jittery scan windows, empty or thin time windows, zero or stalled stamps and a stalled simulation clock, names a failing TF lookup, and turns ERROR when input or integration stops; Table 14 gives the exact levels and the check behind each. With time windows the period is `period_windows` × `window_period_s` seconds; a deployment should measure the rates on the robot, keep a window long enough to hold several messages, and choose `period_windows` from the period it wants to detect. The experiments integrate every message and are unaffected.

Design trade-offs. Three simplifications are deliberate. A single global `survival_decay` replaces the per-cell survival posterior of Rosen et al. (2016) and the learned transition rates of Tipaldi et al. (2013) and Saarinen et al. (2012); E3 shows that, once paired with hysteresis, one uniform rate already yields F1 1.000 at both λ=0.90 and 1.0, but a scene with sharply heterogeneous per-cell dynamics is untested. The flat hash trades memory scaling for simplicity, as above. And two backends behind an `if/else` factory is the right amount of mechanism for two, but a third backend would argue for a registry such as `pluginlib` (Open Robotics, 2024); this is a design judgement, not a measured result.

When not to use STRATA. It is not a substitute for SLAM: it estimates no pose and closes no loops, so a deployment without an external localizer publishing the `map`→sensor transform has no path to a map. It is not multi-session mapping: comparing Monday's map with Friday's or replaying a past session needs the versioning of LT-mapper or Yang et al. (2024). It has no object-level reasoning, unlike the object-level change reasoning of Khronos (Schmid et al., 2024). And it does not discover periods: where the relevant cycles are unknown or are not harmonics of one base period, FreMEn's selection among candidate periods (Krajník et al., 2017) or warped hypertime (Krajník et al., 2019) is the better tool. Within those bounds—one continuous session behind an external localizer, at the raw cell level, in 2D or 3D—it is the intended fit.

Limitations. All evidence is synthetic: every number comes from a deterministic seeded harness driving hand-authored hit and miss patterns through the shipped engine, with no real sensor, robot motion or field deployment, whereas ELite, LT-mapper, Berrio et al. (2022) and the FreMEn grid of Krajník et al. (2016) report long-term real-world evidence. E1–E4 are single-seed and carry no dispersion estimate; only E5 and E6 are multi-seed. No external baseline is run, not even a FreMEn

model driven by the same streams, so this paper does not show that the calibrated test labels periodic cells better than FreMEn's amplitude-ranked component selection, or that the classifier is useful on real sensor data. The evidence establishes a transparent state machine on synthetic streams, not a reliable four-class scene classifier: most aperiodic cells of moderate occupancy end Static, low-duty doors are lost to pruning under the E2 parameters, and at the shipped period correlated clutter is labelled Periodic (below). E4 reports no latency: its timing recordings establish only which backend is cheaper per call. The ROS node has not been run on a robot or on a recorded bag: its evidence is the unit tests of its ROS-free helpers and the synthetic scenarios of Table 14, which continuous integration runs on ROS 2 Humble only, in one container image; the zero-stamp rule, the no-window-closing warning and the sensor-frame cap and expiry have no dedicated check. The saved map was loaded by `nav2_map_server` in that Humble job only; no other distribution is in the continuous integration, and the node does not verify a saved pair itself. The engine models single-session persistence only, forgets only what it re-observes (Remark 3), and loses low-duty-cycle periodic cells to the prune/maturity race of Proposition 5. The periodicity test has limits of its own. What is proved (Propositions 3 and 4) is a per-history bound under an iid null and occupancy-independent touches: at the shipped level, a cell that is never pruned is labelled Periodic at some window of the run with probability at most 0.2. The 0.01 target is a measurement: at most 2 of 2000 clutter cells within 1024 windows on held-out seeds (E5), and for pruned clutter the proved bound grows with the number of re-created histories. The Wilson limits quoted with it are computed per configuration and are not adjusted for the 36 held-out clutter configurations examined. Temporally correlated clutter violates the null, and at the shipped T=24 the test labels most slowly switching cells Periodic (1908 of 2000 in E5). Spending costs detection delay: a noiseless 50%-duty door with T=8 needs 25 windows instead of 15, and one with 10% flipped windows 48 instead of 26. Finally, the dominant labelling error is not periodic at all. The graduation rule labels a cell Static when it is occupied in more than roughly $m^{\dagger}$ of its touched windows (Remark 1; 0.354 with the shipped defaults), so it cannot separate random clutter from a noisy wall of similar occupancy: E3 scores Bernoulli(0.6) cells as walls and E5 scores Bernoulli(0.5) cells as clutter, and with the shipped defaults 1998 and 1974 of 2000 such cells are Static after 256 windows (E6). The defaults were not changed in response; this is an open defect of the default parameters, and E6 covers only iid cells touched in every window under four parameter sets. The natural next steps follow directly: a null that allows temporal correlation, for example by testing the harmonic against a local noise floor rather than against white noise, and an anytime-valid test that also covers re-created histories, for which time-uniform supermartingale bounds (Howard et al., 2020) are the natural tool; a FreMEn baseline on the same synthetic streams; default increments and a graduation rule that reject clutter of moderate occupancy; validation on at least one real 2D and one real 3D LiDAR sequence, and then in the field on a 2D wheeled platform and a 3D-LiDAR platform; per-cell learned decay; exempting recently active cells from pruning, or feeding a running amplitude estimate into (16); and relaxing untouched cells toward the prior on a tick basis, without changing the touched-cell path that E1–E4 characterise.

6. Conclusion

STRATA packages known ingredients of lifelong occupancy mapping—windowed log-odds with survival decay, a Schmitt-trigger graduation band, negative evidence from free-space rays and a FreMEn-lite amplitude test at one configured period, with a significance bound obtained from standard concentration inequalities—into one geometry-free engine that drives a 2D occupancy grid and a 3D voxel hash through a single interface and a one-parameter switch. Only the point-to-key map and the ray walk differ per backend; the classifier that assigns Static, Periodic, Transient or Unknown is written once and builds without ROS. A seeded synthetic characterisation shows near-identical

static-layer quality across backends up to a sampling-induced precision gap, detection of 50%-duty periodicity from 25 windows on, with the probability of ever labelling iid clutter Periodic during an unpruned history bounded by the spent level and measured at no more than 2 of 2000 cells over 1024 windows on held-out seeds, a provable failure mode at low duty cycle, a dominant false Static label on aperiodic cells of moderate occupancy (1960 of 2000 Bernoulli(0.5) cells with the shipped defaults) that the graduation rule cannot avoid, walls that stay Static with periodicity on once the Fourier coefficients are centred, hysteresis as the dominant stabiliser, and a flat estimated per-cell footprint with periodicity off, with `voxel3d` holding 4.4–10.0× as many live cells as `grid2d`. Its closest prior work, the FreMEn spatio-temporal grid of Krajník et al. (2016), has long-term real-world evidence that STRATA lacks; a FreMEn baseline on the same streams and real LiDAR sequences are the next evaluation steps. Within its stated bounds—synthetic evidence, no external baseline, a single session, a fixed base period, an external localizer—STRATA is offered as a small module for other systems to build on.

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A. Reproduction

STRATA is released under the Apache-2.0 license at https://github.com/kjungmo/strata and targets ROS 2 Humble.

Code pin. The revision of that repository that carries this paper, its result files and its reproduction scripts is the tag `paper-v1`. The engine (`strata_core`), the node (`strata`) and the check scripts (`scripts/`) in it are those of commit `312b09c` (11 October 2026): the experiment harness, experiment E6, the result files and this manuscript were completed after that commit without changing those three directories, and the number check compares their tree identifiers with that commit's. Commit `312b09c` itself predates E6 and the reproduction scripts named below.

What reproduces. `paper/experiments/check_reproduction.sh` rebuilds the harness from a clean checkout and requires every result file of E0–E3, E5 and E6 to come out byte for byte; `paper/experiments/reproduction_log.txt` lists the compilers with which it did (GCC 9.4.0 with libstdc++, on one x86-64 Linux host). The result is not independent of the C++ standard library. The streams come from `std::mt19937`, whose output the standard fixes, through three standard distributions, whose algorithms it does not fix. With clang 23.1.2 and libc++ (`check_stdlib.sh`, record in `results/stdlib_libcxx/`), 15 of the 19 seeded result files are byte-identical and four differ: the two E1 files, the E0 door file and the E6 confusion file. These are the experiments whose outcome depends on integers drawn with `std::uniform_int_distribution` (mover cells in E1's `grid2d` runs, door phases in E0 and E6). Under libc++ the final `grid2d` precision at high clutter in E1 is 0.9583 with 168 predicted cells instead of 0.9253 with 174, and every other final value of

Table 1 is unchanged; five of the 20 E0 door medians move by one window (at δ=0.1, 20 to 21 and 23 to 24) and the selected levels do not change; and 1976 instead of 1960 of the 2000 aperiodic cells of E6 are Static with the shipped defaults. E5's confusion draws door phases the same way, but its counts do not depend on the phases and its file is identical. No third standard library was tried.

From the repository root:

- Core engine and unit tests, no ROS required (continuous integration runs the same build with g++, Eigen3 and GoogleTest on `ubuntu-latest`):

  ```
  cmake -S strata_core -B build -DSTRATA_CORE_BUILD_TESTS=ON
  cmake --build build && (cd build && ctest)
  ```

- Unit tests, the calibration E0 (seeds from 20260928), experiments E1–E4 (seed 12345), E5 (calibration seeds from 30260928, evaluation seeds from 40260928) and E6 (sweep seeds from 50260928), writing the CSVs under `paper/experiments/results/`. E0–E3, E5 and E6 rewrite their files byte for byte. E4 reads the wall clock: this script writes its output to `paper/experiments/build/e4_throughput_local.csv` and fails if a cell count or byte estimate differs from the committed one (`check_e4_local.py`); timings for the paper are recorded separately with `paper/experiments/measure_e4.sh`. The comparison columns of Table 3 are the same harness built against the v0.1.0 `strata_core` (`results/pre_fix_2026-09-28/`), against the centred engine before the significance test (`results/pre_calibration_2026-09-28/`) and against the engine with the single-read-out level (`results/pre_spending_2026-09-28/`):

  ```
  bash paper/experiments/run_all.sh
  ```

- Tables and plot data of this paper, and the number check (`paper/arxiv_ref/NUMBERS.md`). The check regenerates every table and plot-data file and compares it with the committed one. For the hand-written text it locates each numeral and requires it to be tied to a value recomputed from the result files or the code, or to be listed as structural with a reason (an experiment design constant restated from the harness, a symbol in a formula, a plot layout value). A numeral that is neither fails the check. Its limits are these. Structural numerals, among them the scenario sizes of §4.1 and the design constants written as symbol and value, are not parsed from the harness; they are kept in a committed list (`paper/arxiv_ref/structural_numerals.tsv`), so that changing one fails the check until the list is rewritten on purpose. A numeral that is tied to a recomputed value only by being equal to it, and not by its sentence, could be exchanged for another recomputed value without the check noticing; `NUMBERS.md` gives the count of each kind.

  ```
  python3 paper/arxiv_ref/tools/make_artifacts.py
  python3 paper/experiments/number_guard.py
  ```

Alternative backends can be replayed against the reference scenario of the unit test `Integration.WallStaticMoverTransientDoorPeriodic`.

B. Parameters

Table 11: Engine parameters and shipped defaults (the `LayeredMapParams` struct and both `params/*.yaml` files agree). Geometry and node parameters are not part of the engine: `grid_width/grid_height` 400, `grid_resolution` 0.05 m and origin $(-10, -10)$ m for `grid2d`; `voxel_size` 0.2 m for `voxel3d`; `publish_period` 1.0 s; `window_mode time` in both files (`scans` when the parameter is not set) with `window_period_s` 1.0 s; `expected_scan_rate_hz` 0 (estimated from stamps), `rate_warn_drop_fraction` 0.05, `input_timeout_s` 5 s and `startup_timeout_s` 30 s.

Parameter	Symbol	Default	Meaning
`layer_interval`	L	10	integration ticks per window
`l_hit`	l_{hit}	0.85	log-odds increment on an occupied window
`l_miss`	l_{miss}	−0.4	log-odds increment on a free window
`l_min`, `l_max`	$l_{\min}, l_{\max}$	−5.0, 5.0	clamp bounds
`survival_decay`	λ	0.97	per-window forgetting multiplier
`graduate_prob`	p_{grad}	0.8	probability to graduate to Static
`demote_prob`	p_{dem}	0.45	probability at or below which Static demotes
`min_observations`	$N_{\min}$	3	touched windows before graduation
`prune_prob`	p_{prune}	0.05	erase non-static, non-periodic cells below this
`enable_periodicity`	—	true	run the Fourier model
`periodic_amplitude_min`	$a_{\min}$	0.3	minimum harmonic amplitude (effect size)
`periodic_false_alarm`	δ	0.2	level of the significance test
`periodic_alpha_spending`	δ_n	true	spend δ over the touch count, Eq. (15)
`period_windows`	T	24	base period and amplitude gate
`n_harmonics`	H	2	harmonics per cell

Table 12: Per-experiment deviations from the defaults of Table 11, as compiled into each harness. "—": default; "swept": varied over the E3 grid; "n/a": inert because periodicity is off; "false / true": run once each way, periodicity on at the defaults.

Parameter (default)	E1	E2	E3	E4
L (10)	1	1	1	1
`enable_periodicity` (true)	false / true	true	false / true	false
l_{hit} (0.85)	—	1.0	—	—
l_{miss} (−0.4)	—	−1.0	—	—
λ (0.97)	—	1.0	swept	1.0
p_{grad} (0.8)	—	0.9	swept	—
p_{dem} (0.45)	—	0.4	swept	—
$N_{\min}$ (3)	—	5	—	—
p_{prune} (0.05)	—	—	0.01	—
T (24)	—	8	—	n/a
H (2)	—	3	—	n/a
δ (0.2, spent)	—	—	—	n/a

C. Interfaces and State Transitions

Table 13: ROS 2 input/output contract of the `strata` node. Every scan or cloud is transformed with a full 6-DoF lookup of `global_frame` (default `map`) ← sensor frame at the message stamp; a lookup that still fails after the wait of Table 14 drops the message with a throttled warning. The node neither subscribes to `/initialpose` nor broadcasts `map`→`odom`; one mutex serialises integration, publishing, saving and the diagnostics status. Both timers run on the wall clock.

Direction	Name (default)	Type, QoS	Condition
Subscribe	`scan_topic` (`/scan`)	`LaserScan`, sensor data	`grid2d`
Subscribe	`points_topic` (`/points`)	`PointCloud2`, sensor data	`voxel3d`
Publish	`~/map`	`OccupancyGrid`, depth 1, transient local	`grid2d`
Publish	`~/map_points`	`PointCloud2`, depth 1, volatile	`voxel3d`
Service	`~/save_map`	`Trigger`	both (`.pgm`+`.yaml` / `.pcd`)
Timer	`publish_period` (1.0 s)	—	both
Publish	`/diagnostics`	`DiagnosticArray`, depth 10	both, one status a second

Node behaviour that a deployment depends on. The node's rules for time, transforms, input health and saving live in ROS-free headers with their own unit tests, and continuous integration runs the launched node against synthetic input on ROS 2 Humble. Table 14 states each rule as implemented and names the check that exercises it; a rule without a check says so. None of these checks uses a robot or a recorded bag (§5).

Table 14: Behaviour of the `strata` node at the pinned commit and the committed check that exercises each rule. Unit tests are the 64 gtest cases of the node package (`strata/test/`); scripts are in `scripts/` and run in the Humble job of the continuous integration against the launched node with synthetic input. "—": no dedicated check.

Topic	Rule as implemented	Checked by
Time windows	Windows of `window_period_s` on header stamps, in integer nanoseconds from the first stamp. A later stamp closes the open window and every silent one. A stamp more than half a window behind the newest, or more than 10^6 windows ahead, re-anchors the clock after closing the open window.	`test_window_clock` (6 cases); `synthetic_e2e.py` with time windows
Zero stamps	Dropped with time windows, integrated with scan windows; reported as a warning in both.	—
TF wait	The lookup at the message stamp is polled without blocking for 0.1 s of ROS time, at most 1 s of steady time. A ROS clock unchanged for 1 s counts as frozen and is not waited on. A clock that steps past the budget gets 20 ms more. A failed lookup drops the message.	`test_tf_wait` (7 cases); `check_coarse_clock.py`; `check_sim_clock_stall.py`
Input status	One status a second on `/diagnostics`. No message yet: OK, then ERROR after `startup_timeout_s`. Messages stop: ERROR after `input_timeout_s`. Lookups fail before anything was integrated: OK with the failing lookup named, then WARN after `startup_timeout_s`. Lookups fail later: named once nothing was integrated for 1 s, then ERROR after `input_timeout_s`. Every status carries message, integration and TF-failure totals, the running failure streak and the last tf2 error text.	`test_input_status` (18 cases); `check_no_localizer.py` on both backends
Clock and stamp warnings	WARN when `use_sim_time` is set and the ROS clock stands still for 3 s while messages arrive, and when messages are integrated but no time window closes for 10 window periods (and more than 2 s).	`check_sim_clock_stall.py` (clock); — (no window closing)
Rate statistics	Per sensor frame, from the stamps of integrated messages: rate, window duration, and loss estimated from gaps against the 20th-percentile interval or `expected_scan_rate_hz`. At most 16 frames are tracked; a frame silent for 10 window closes is forgotten. WARN on loss above `rate_warn_drop_fraction` or jittery scan windows, and on empty or thin time windows.	`test_rate_monitor` (4 cases); `synthetic_e2e.py` with 40 % of the scans withheld; — (frame cap and expiry)
Saving	`grid2d`: PGM and `nav2_map_server` YAML are written to temporary files, flushed, renamed (image first) and the directory flushed. A save that fails before the renames leaves the previous pair untouched and answers `success: false` with the reason. A failure between the two renames can leave the new image beside the previous YAML, so the YAML records the size and SHA-256 of its image and `check_saved_map.py` verifies a pair; neither the node nor Nav2 runs that check. `voxel3d`: one PCD through the same steps.	`test_map_writer` (25 cases); saved-pair step of the CI job; `synthetic_e2e.py` with `--check-map-server`
Saved-map values	Free 254, Transient and Periodic 100, Static 0, Unknown 205 with thresholds 0.65 and 0.196, so `nav2_map_server` serves free, unknown, occupied and unknown.	`synthetic_e2e.py` with `--check-map-server`: all 160,000 cells of the synthetic scene, which holds no Transient cell
Parameters	Every key of both shipped YAML files is declared by the node and holds the file's value.	`check_param_binding.py` on both backends

Table 15: Cell-class transitions, all evaluated at window close.

From	To	Trigger
Unknown (absent)	Transient	first hit or miss creates the cell: ℓ=0, p=0.5 $\geq p_{\text{prune}}$
Transient	Static	Eq. (7): $p \geq p_{\text{grad}}$, $N \geq N_{\text{min}}$, $\neg\pi$
Transient	Periodic	π, Eq. (13): $n \geq T$ and, for some k, $a_k \geq a_{\text{min}}$ and $H\,B(d_k) \leq \delta_n$
Transient	Unknown (erased)	Eq. (16): $p < p_{\text{prune}}$, $a < a_{\text{min}}$, $\neg g$
Static	Transient / Periodic / Unknown	Eq. (8), then re-read by Eq. (17)
Periodic	Transient / Unknown	π becomes false (pruned only once also $a < a_{\text{min}}$)
Periodic	Static	$\neg\pi$ and Eq. (7) holds
Periodic	—	never graduates while π; never pruned while $a \geq a_{\text{min}}$
Static	—	never pruned while g

D. Extended Results

Table 16: Full E3 sweep (36 configurations, 50 noisy wall cells, occupancy probability 0.6, 10 movers/window, 80 windows, seed 12345). Deg. marks the zero-band configuration $p_{\text{dem}} = p_{\text{grad}}$. F1 to Flicker: periodicity off. *Flicker, P on*: the same sweep with periodicity on at the shipped T=24, H=2, $a_{\min}$=0.3, δ=0.2 spent over the touch count (current engine; its F1 equals the periodicity-off F1 in 36 of 36 rows).

p_{grad}	p_{dem}	λ	Band	Deg.	F1	Precision	Recall	Flicker	Flicker, P on
0.6	0.3	0.90	0.3	no	1.000	1.000	1.000	58	58
0.6	0.5	0.90	0.1	no	1.000	1.000	1.000	108	108
0.6	0.6	0.90	0.0	yes	1.000	1.000	1.000	170	170
0.6	0.3	0.97	0.3	no	1.000	1.000	1.000	54	54
0.6	0.5	0.97	0.1	no	1.000	1.000	1.000	72	72
0.6	0.6	0.97	0.0	yes	1.000	1.000	1.000	92	92
0.6	0.3	1.00	0.3	no	1.000	1.000	1.000	52	52
0.6	0.5	1.00	0.1	no	1.000	1.000	1.000	68	68
0.6	0.6	1.00	0.0	yes	1.000	1.000	1.000	76	76
0.7	0.3	0.90	0.4	no	1.000	1.000	1.000	56	56
0.7	0.5	0.90	0.2	no	1.000	1.000	1.000	94	94
0.7	0.7	0.90	0.0	yes	0.980	1.000	0.960	250	250
0.7	0.3	0.97	0.4	no	1.000	1.000	1.000	54	54
0.7	0.5	0.97	0.2	no	1.000	1.000	1.000	66	66
0.7	0.7	0.97	0.0	yes	1.000	1.000	1.000	116	116
0.7	0.3	1.00	0.4	no	1.000	1.000	1.000	52	52
0.7	0.5	1.00	0.2	no	1.000	1.000	1.000	62	62
0.7	0.7	1.00	0.0	yes	1.000	1.000	1.000	86	86
0.8	0.3	0.90	0.5	no	1.000	1.000	1.000	56	56
0.8	0.5	0.90	0.3	no	1.000	1.000	1.000	82	82
0.8	0.8	0.90	0.0	yes	0.947	1.000	0.900	343	343
0.8	0.3	0.97	0.5	no	1.000	1.000	1.000	52	52
0.8	0.5	0.97	0.3	no	1.000	1.000	1.000	56	56
0.8	0.8	0.97	0.0	yes	1.000	1.000	1.000	124	124
0.8	0.3	1.00	0.5	no	1.000	1.000	1.000	52	52
0.8	0.5	1.00	0.3	no	1.000	1.000	1.000	56	56
0.8	0.8	1.00	0.0	yes	1.000	1.000	1.000	84	84
0.9	0.3	0.90	0.6	no	1.000	1.000	1.000	54	54
0.9	0.5	0.90	0.4	no	0.990	1.000	0.980	71	71
0.9	0.9	0.90	0.0	yes	0.810	1.000	0.680	574	574
0.9	0.3	0.97	0.6	no	1.000	1.000	1.000	52	52
0.9	0.5	0.97	0.4	no	1.000	1.000	1.000	54	54
0.9	0.9	0.97	0.0	yes	1.000	1.000	1.000	150	150
0.9	0.3	1.00	0.6	no	1.000	1.000	1.000	52	52
0.9	0.5	1.00	0.4	no	1.000	1.000	1.000	52	52
0.9	0.9	1.00	0.0	yes	1.000	1.000	1.000	122	122

E. Notation

Table 17: Symbols used in §3.

Symbol	Meaning
C, i	set of integer cell ids; a cell
κ_b, ρ_b	backend key map and free-space ray walk
Γ_b, Φ, Ψ	evidence extractor, engine update, class read-out (Eq. 1)
$h_{i,w}$, $m_{i,w}$	hit and miss counts of cell i in window w
τ_i, o_i, f_i	touched, occupied and free indicators of a window
ℓ_i, p_i	log-odds occupancy and $\sigma(\ell_i)$
N_i, n_i	touched-window counts for graduation and for the Fourier model
g_i, π_i	graduated flag and periodic predicate
a_i, $a_{i,k}$, α_k, β_k	dominant and harmonic amplitudes, harmonic coefficients
$d_{i,k}$, $\mathbf{M}_{i,k}$, B	explained variance, centred phase design, Chernoff tail bound (Eq. 12)
δ, $r^\star$, $a^\star(n)$	nominal level, its critical d, amplitude threshold at uniform coverage (Eq. 14)
δ_n, L_N	level spent at touch count n (Eq. 15); histories begun by window N (Prop. 4)
λ, l_{hit}, l_{miss}	survival decay and log-odds increments
p_{grad}, p_{dem}, p_{prune}	graduation, demotion and prune thresholds
T, H, ω, $a_{\min}$	base period, harmonics, base frequency, amplitude threshold

F. Trajectory-Level Results in Full

Table 18: All E5 held-out configurations: cells (of 2000) labelled Periodic at some window up to N=1024, and in parentheses the mean number of histories a cell began (pruning and re-creation). Columns: parameter set, touch probability q, rule (S: single read-out, δ=0.1; Sp: spent, δ=0.2).

	E2 parameters (T=8, H=3)				Shipped (T=24, H=2)			
Cell	q=1, S	q=1, Sp	q=0.5, S	q=0.5, Sp	q=1, S	q=1, Sp	q=0.5, S	q=0.5, Sp
Bernoulli m=0.1	3 (266)	0 (266)	3 (133)	0 (133)	1 (76)	0 (76)	0 (38)	0 (38)
Bernoulli m=0.2	141 (180)	0 (180)	77 (91)	0 (91)	77 (41)	0 (41)	40 (21)	0 (21)
Bernoulli m=0.3	620 (112)	0 (112)	330 (57)	0 (57)	202 (12)	0 (12)	102 (7)	1 (7)
Bernoulli m=0.4	953 (62)	2 (62)	543 (32)	2 (32)	88 (2)	2 (2)	58 (2)	0 (2)
Bernoulli m=0.5	717 (23)	2 (23)	350 (12)	1 (12)	67 (1)	0 (1)	50 (1)	1 (1)
Bernoulli m=0.6	199 (5)	0 (5)	96 (3)	0 (3)	42 (1)	0 (1)	52 (1)	1 (1)
Bernoulli m=0.7	16 (1)	0 (1)	12 (1)	0 (1)	9 (1)	0 (1)	18 (1)	0 (1)
Bernoulli m=0.8	4 (1)	0 (1)	6 (1)	0 (1)	2 (1)	0 (1)	4 (1)	0 (1)
Bernoulli m=0.9	0 (1)	0 (1)	0 (1)	0 (1)	1 (1)	0 (1)	0 (1)	0 (1)
always occupied	0 (1)	0 (1)	0 (1)	0 (1)	0 (1)	0 (1)	0 (1)	0 (1)
always free	0 (342)	0 (342)	0 (171)	0 (171)	0 (114)	0 (114)	0 (57)	0 (57)
Markov (correlated)	349 (98)	2 (98)	79 (37)	0 (37)	1998 (15)	1908 (15)	1331 (5)	409 (5)